FLUID MECHANICS

Subject Code: 23A01303

UNIT - 1: Basic Concepts and Definitions

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Distinction between a fluid and a solid; Density, Specific weight, Specific gravity, Kinematic and dynamic viscosity; Variation of viscosity with temperature, Newton's law of viscosity; Vapor pressure, Boiling point, Surface tension, Capillarity, Bulk modulus of elasticity, Compressibility.

1. Introduction

Fluid Mechanics

It is the branch of science that deals with the behaviour of fluids (liquids and gases) at rest as well as in motion, and the forces acting on them. Its three branches are: Fluid Statics (fluids at rest), Fluid Kinematics (motion without considering forces) and Fluid Dynamics (motion with the forces causing it).

Fluid

A fluid is a substance that deforms continuously when even a very small shear force is applied to it, however small the force may be. Liquids and gases are fluids. Fluids have no definite shape; they take the shape of the container.

Distinction between a Solid and a Fluid

Property	Solid	Fluid
Shear force	Resists by a fixed deformation (angle) and then stops	Deforms continuously as long as the force acts
Shape	Definite shape and volume	No definite shape; takes shape of container
Molecular spacing	Very close, strong cohesive forces	Larger spacing, weak cohesive forces
Stress	Depends on strain (Hooke's law)	Depends on rate of strain (Newton's law)
Tensile strength	Can withstand tension, compression, shear	Can withstand only compression; almost no tension
Examples	Steel, wood, concrete	Water, oil, air, mercury

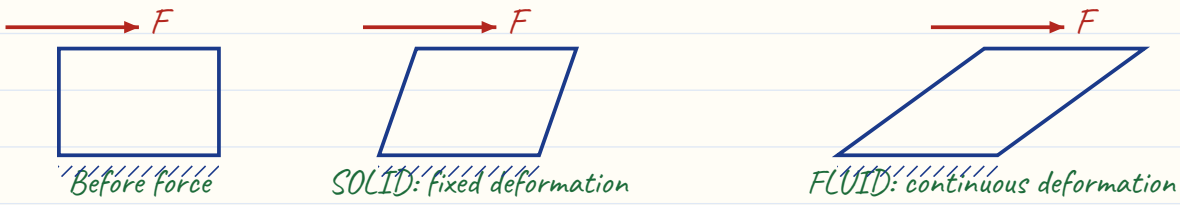


Fig 1.1: Response of a solid and a fluid to a shear force

Types of Fluids

- Ideal fluid – incompressible and has zero viscosity (imaginary, no shear stress). It is used for simplifying analysis.
- Real fluid – has viscosity (all actual fluids). Shear stress exists when there is relative motion.
- Newtonian fluid – obeys Newton's law of viscosity, $\tau = \mu (du/dy)$. Eg: water, air, kerosene, alcohol.
- Non-Newtonian fluid – does not obey the linear law. Eg: blood, paint, toothpaste, slurries, polymer solutions.
- Ideal plastic – shear stress exceeds a yield value, then remains constant for any strain rate.
- Compressible fluid – density changes with pressure (gases); Incompressible fluid – density is constant (liquids).

Continuum Concept

Fluid is made of molecules, but in engineering we assume it is a continuous medium (continuum): properties like density, pressure and velocity vary smoothly from point to point. This is valid when the characteristic length of the problem is much larger than the mean free path of molecules (Knudsen number is small).

Units and Dimensions (SI)

Quantity	Symbol	SI Unit	Dimension
Mass	m	kg	M
Length	L	m	L
Time	t	s	T
Force	F	$N = kg\cdot m/s^2$	$M L T^{-2}$
Pressure	p	$Pa = N/m^2$	$M L^{-1} T^{-2}$
Density	ρ	kg/m^3	$M L^{-3}$
Specific weight	w	N/m^3	$M L^{-2} T^{-2}$
Dynamic viscosity	μ	$N\cdot s/m^2 = Pa\cdot s$	$M L^{-1} T^{-1}$
Kinematic viscosity	ν	m^2/s	$L^2 T^{-1}$
Surface tension	σ	N/m	$M T^{-2}$
Bulk modulus	K	N/m^2	$M L^{-1} T^{-2}$

2. Basic Fluid Properties

2.1 Density (Mass Density), ρ

Mass of the fluid per unit volume.

$$\rho = m / V \ (kg/m^3)$$

- Water at 4°C: $\rho = 1000\ kg/m^3$; Sea water ≈ $1025\ kg/m^3$; Mercury = $13600\ kg/m^3$; Air at 15°C, 1 atm = $1.225\ kg/m^3$.
- Density of a liquid decreases slightly with temperature; for gases $\rho = p/(RT)$ (equation of state, $R = 287\ J/kg\cdot K$ for air).

2.2 Specific Weight (Weight Density), w or γ

Weight of the fluid per unit volume.

$$w = W / V = \rho\, g \ (N/m^3)$$

- Water: $w = 1000 \times 9.81 = 9810\ N/m^3$ ($9.81\ kN/m^3$). Air: $w = 1.225 \times 9.81 \approx 12.01\ N/m^3$.
- Unlike density, specific weight changes with location because g changes.

2.3 Specific Volume, v

Volume of fluid per unit mass. $v = V/m = 1/\rho$ (m^3/kg). Used mostly for gases.

2.4 Specific Gravity (Relative Density), S

Ratio of density (or specific weight) of the fluid to the density (specific weight) of a standard fluid. Standard fluid = water at 4°C for liquids; air (or hydrogen) at standard conditions for gases.

$$S = \rho_{fluid} / \rho_{water} = w_{fluid} / w_{water} \text{ (no units)}$$

- Mercury $S = 13.6$; Sea water ≈ 1.025; Petrol ≈ 0.72; Glycerin $= 1.26$; Kerosene ≈ 0.81.
- So $\rho = S \times 1000$ kg/m³ and $w = S \times 9810$ N/m³.

Example 1

A liquid of specific gravity 0.8 occupies 2 m³. Find (a) density, (b) mass, (c) weight, (d) specific weight.

Sol: (a) $\rho = S \times 1000 = 0.8 \times 1000 = 800$ kg/m³

(b) $m = \rho V = 800 \times 2 = 1600$ kg

(c) $W = mg = 1600 \times 9.81 = 15696$ N ≈ 15.7 kN

(d) $w = \rho g = 800 \times 9.81 = 7848$ N/m³

Fluid (20°C)	ρ (kg/m³)	S	μ (Pa·s)
Water	998	0.998	1.0×10^{-3}
Air	1.2	0.0012	1.8×10^{-5}
Mercury	13600	13.6	1.55×10^{-3}
Glycerin	1260	1.26	1.49
SAE 30 oil	917	0.917	0.29

3. Viscosity

Viscosity

It is the property of a fluid by virtue of which it offers resistance to the relative motion (shear deformation) between its adjacent layers. It is due to cohesion between molecules (liquids) and molecular momentum exchange (gases). It is the 'internal friction' of the fluid.

Newton's Law of Viscosity

Consider a fluid between two parallel plates, the lower plate fixed and the upper one moving with a velocity U under a force F. Fluid sticks to the plates (no-slip condition), so velocity varies from 0 at the fixed plate to U at the moving plate.

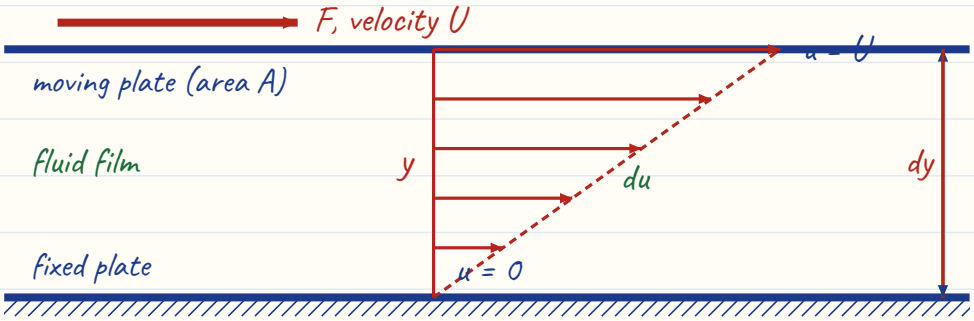


Fig 1.2: Velocity profile between two parallel plates

Newton's law states that the shear stress (τ) between adjacent fluid layers is directly proportional to the velocity gradient (rate of shear strain):

$$\tau = F/A = \mu \, (du/dy)$$

- τ = shear stress (N/m^2); du/dy = velocity gradient = rate of angular deformation (s^{-1}); μ = constant of proportionality = coefficient of dynamic viscosity.
- For linear profile: $du/dy = U/y$, so $\tau = \mu U / y$ and $F = \mu A U / y$.
- Fluids which obey this law are Newtonian fluids; the τ–du/dy graph is a straight line through the origin with slope μ.

Dynamic Viscosity (Absolute Viscosity), μ

$$\mu = \tau \, / \, (du/dy)$$

- SI unit: $N{\cdot}s/m^2 = Pa{\cdot}s = kg/(m{\cdot}s)$. CGS unit: poise ($dyne{\cdot}s/cm^2 = g/(cm{\cdot}s)$).
- 1 Pa·s = 10 poise; 1 poise = 0.1 $N{\cdot}s/m^2$; 1 centipoise = 0.001 Pa·s (water at 20°C ≈ 1 cP).
- Dimensions: $M\,L^{-1}\,T^{-1}$.

Kinematic Viscosity, ν

Ratio of dynamic viscosity to mass density. It is called 'kinematic' since force is not involved in its dimensions.

$$\nu = \mu \, / \, \rho$$

- SI unit: m^2/s. CGS unit: stoke (cm^2/s). 1 stoke = 10^{-4} m^2/s; 1 m^2/s = 10^4 stokes; 1 centistoke = 10^{-6} m^2/s.

Basic Concepts and Definitions

- Dimensions: $L^2 T^{-1}$. Water at 20°C: $\nu \approx 1 \times 10^{-6}$ m²/s (1 cSt); air: $\nu \approx 1.5 \times 10^{-5}$ m²/s.

Note: Water is 'thicker' (larger μ) than air, but air has a larger kinematic viscosity because its density is very small.

Types of Fluids on the Basis of Newton's Law

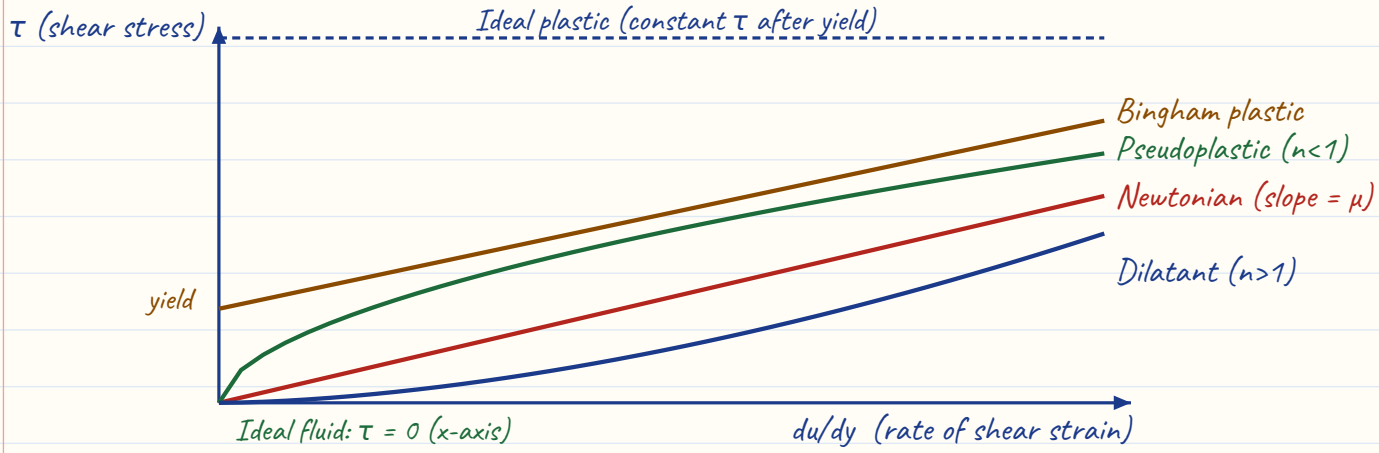


Fig 1.3: Rheological diagram (shear stress vs rate of shear strain)

- Newtonian: $\tau = \mu (du/dy)$. Constant μ. Water, air, oils, gasoline.
- Non-Newtonian: $\tau = A (du/dy)^n + B$. $n < 1$ → pseudoplastic (blood, paint, polymer solution; apparent viscosity falls as shear increases); $n > 1$ → dilatant (quicksand, starch suspension; viscosity rises with shear).
- Bingham plastic: needs a minimum yield stress before it flows, then behaves linearly (toothpaste, sewage sludge, mud).
- Ideal fluid: $\tau = 0$ for any du/dy. Elastic solid: $du/dy = 0$ for any τ (the y-axis).

Variation of Viscosity with Temperature

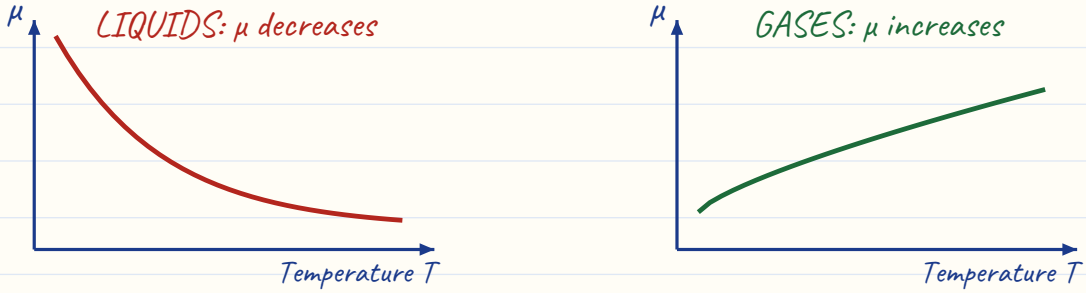


Fig 1.4: Effect of temperature on viscosity

- Liquids: viscosity decreases as temperature increases. Reason: viscosity is due to cohesive forces between molecules. On heating, molecules gain energy and cohesion weakens. (Engine oil flows easily when hot.)
- Gases: viscosity increases as temperature increases. Reason: viscosity is due to molecular momentum transfer between layers. On heating, molecular agitation rises, the exchange of momentum increases and so does the resistance.

Basic Concepts and Definitions

- Liquids – Andrade's equation: $\mu = A\, e^{B/T}$ or $\mu = \mu_0 / (1 + \alpha t + \beta t^2)$ (t in °C; α, β are constants).
- Gases – Sutherland's law: $\mu = C\, T^{3/2} / (T + S)$ (S = Sutherland constant, ≈ 110.4 K for air).
- Pressure has negligible effect on viscosity except at very high pressures.

Solved Problems on Viscosity

Example 2 – Flat plate over a fixed plate

The space between two parallel plates, 2.5 mm apart, is filled with oil of viscosity 0.9 N·s/m². The upper plate moves at 1.5 m/s. Find the shear stress in the oil. If plate area is 0.4 m², find the force needed.

Sol: $du/dy = U/y = 1.5 / 0.0025 = 600\ s^{-1}$

$\tau = \mu\, du/dy = 0.9 \times 600 = 540\ N/m^2$

$F = \tau\, A = 540 \times 0.4 = 216\ N$

Example 3 – Unit conversion

An oil of specific gravity 0.9 has a dynamic viscosity of 6 poise. Find its kinematic viscosity in stokes.

Sol: $\mu = 6$ poise $= 6 \times 0.1 = 0.6\ N·s/m^2$; $\rho = 0.9 \times 1000 = 900\ kg/m^3$

$\nu = \mu/\rho = 0.6/900 = 6.667 \times 10^{-4}\ m^2/s = 6.667 \times 10^{-4} \times 10\text{^}4 = 6.667$ stokes

Example 4 – Block sliding on an inclined plane

A square block (0.3 m side) weighing 200 N slides down a 20° incline on an oil film of thickness 0.2 mm and $\mu = 0.1\ N·s/m^2$. Find the steady velocity.

Sol: Driving force along plane $= W \sin 20° = 200 \times 0.3420 = 68.4\ N$

At steady speed the viscous force equals this force: $\tau = F/A = 68.4 / (0.3 \times 0.3) = 760\ N/m^2$

$\tau = \mu\, U / y \rightarrow U = \tau\, y / \mu = 760 \times 0.0002 / 0.1 = 1.52\ m/s$

Example 5 – Journal bearing (torque and power)

A shaft of diameter 80 mm rotates at 300 rpm in a bearing of length 60 mm. Radial clearance = 0.2 mm filled with oil, μ = 0.1 Pa·s. Find torque and power lost.

Sol: $U = \pi DN/60 = \pi \times 0.08 \times 300/60 = 1.2566$ m/s

$\tau = \mu U / c = 0.1 \times 1.2566 / 0.0002 = 628.3$ N/m^2

Area $A = \pi D L = \pi \times 0.08 \times 0.06 = 0.01508$ m^2; $F = \tau A = 9.475$ N

Torque $T = F \times D/2 = 9.475 \times 0.04 = 0.379$ N·m

Power $P = T\omega = 0.379 \times (2\pi \times 300/60 = 31.42) = 11.9$ W

4. Vapor Pressure, Boiling Point and Cavitation

Vapor Pressure (p_v)

When a liquid is in a closed vessel, molecules escape from the surface into the space above and exert a pressure on it. At equilibrium (rate of evaporation = rate of condensation) this pressure is the saturation vapor pressure. It increases with temperature.

- Water: $p_v \approx 2.34$ kPa at 20°C; 7.4 kPa at 40°C; 101.3 kPa at 100°C.
- Liquids with high vapor pressure (petrol, alcohol) are called volatile.

Boiling Point

A liquid boils when its vapor pressure becomes equal to the surrounding (absolute) pressure. Hence boiling point decreases when the external pressure is reduced (water boils below 100°C on a mountain; in a pressure cooker boiling point rises above 100°C).

Cavitation

If, in a flowing liquid, the local absolute pressure falls to the vapor pressure, the liquid vaporises and vapor bubbles form. When these bubbles are carried to a high pressure zone they collapse suddenly (implode), producing very high local pressures (hundreds of MPa), noise, vibration, pitting and erosion of metal surfaces, and fall in efficiency.

- Occurs in pump suction, turbine runner blades, propellers, venturi throats, and control valves.
- Prevention: keep pressure above p_v (reduce suction lift, increase NPSH), use smooth surfaces, cavitation-resistant materials, lower speeds.

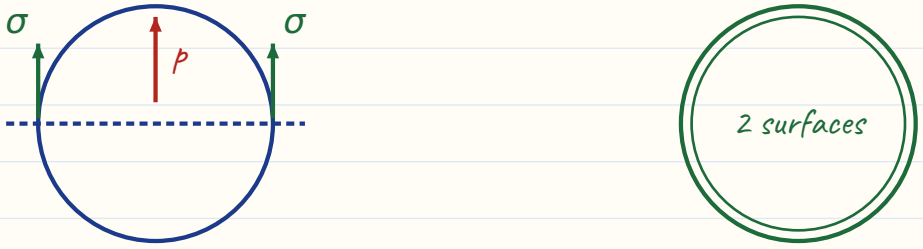
5. Surface Tension

Surface Tension (σ)

Surface tension is the tensile force acting on the surface of a liquid in contact with a gas (or another immiscible liquid) that tends to minimise the surface area. It is the force per unit length of a line drawn on the surface, perpendicular to it. Unit: N/m. Dimension: $M T^{-2}$.

- Cause: Cohesion. A molecule inside the liquid is pulled equally in all directions. A molecule at the surface has a net inward pull, so the surface behaves like a stretched elastic membrane.
- Examples: small droplets are spherical, a needle floats on water, insects walk on water, formation of soap bubbles.
- σ decreases as temperature rises; soaps and detergents reduce σ.
- Values at 20°C: water–air = 0.0728 N/m (0.073); mercury–air = 0.465–0.52 N/m; alcohol–air ≈ 0.022 N/m; soap solution ≈ 0.025 N/m.

Pressure Inside a Droplet, Soap Bubble and Liquid Jet



Droplet: $\Delta p = 4\sigma/d$ Soap bubble: $\Delta p = 8\sigma/d$

Fig 1.5: Forces on a half droplet / bubble

(i) Hollow droplet of diameter d (one free surface). Consider half the droplet. Pressure force (excess pressure Δp over projected area) balances the surface tension force around the circumference:

$$\Delta p \times (\pi/4)\, d^2 = \sigma \times \pi\, d \rightarrow \Delta p = 4\sigma / d$$

(ii) Soap bubble has two surfaces (inner and outer film), so the tension force is doubled:

$$\Delta p \times (\pi/4)\, d^2 = 2 \times \sigma\, \pi\, d \rightarrow \Delta p = 8\sigma / d$$

(iii) Liquid jet of diameter d and length L: $\Delta p \times (d\, L) = 2\sigma\, L \rightarrow \Delta p = 2\sigma / d$

Note: Smaller the droplet, larger the internal pressure. Δp is the excess of internal pressure over the outside pressure (gauge).

Example 6
Find the excess pressure inside (a) a water droplet of diameter 0.5 mm (σ = 0.0736 N/m) and (b) a soap bubble of 40 mm diameter (σ = 0.025 N/m).
Sol: (a) $\Delta p = 4\sigma/d = 4 \times 0.0736 / 0.0005 = 588.8\ N/m^2$
(b) $\Delta p = 8\sigma/d = 8 \times 0.025 / 0.04 = 5\ N/m^2$

6. Capillarity

Capillarity
The phenomenon of rise or fall of a liquid in a small-diameter tube (capillary) relative to the general level of the liquid outside is called capillarity. It occurs due to the combined effect of cohesion (liquid–liquid attraction) and adhesion (liquid–solid attraction).

- If adhesion > cohesion (water–glass), the liquid wets the wall, meniscus is concave, contact angle $\theta < 90°$ → capillary rise.
- If cohesion > adhesion (mercury–glass), meniscus is convex, $\theta > 90°$ ($\approx 130°$–$140°$) → capillary depression.



Fig 1.6: Capillary rise (water) and depression (mercury)

Expression for Capillary Rise / Fall

Let a tube of diameter d be dipped in a liquid of density ρ and surface tension σ, with contact angle θ. Let h be the height of rise.

- Upward force due to surface tension = $\sigma \cos\theta \times$ (circumference) = $\sigma \cos\theta \times \pi d$
- Weight of liquid column raised = $\rho g \times (\pi/4) d^2 \times h$

Equating them (for equilibrium):

$$\sigma \cos\theta\ \pi d = \rho g (\pi/4) d^2 h$$
$$h = 4\sigma \cos\theta / (\rho g d)$$

- For water–clean glass $\theta \approx 0°$, so $h = 4\sigma/(\rho g d)$. For mercury $\theta \approx 130°$, $\cos\theta$ negative → h negative = depression.

- $h \propto 1/d$ – narrower the tube, greater the rise. Capillarity is negligible for $d > 12$ mm.
- Applications: rise of water in soil and plants, ink in blotting paper, oil in a lamp wick. Piezometers/manometer tubes should be > 12 mm bore to avoid capillary errors.

Example 7

Calculate the capillary rise of water ($\sigma = 0.0736$ N/m, $\theta = 0°$) in a 2 mm glass tube. Also find the depression of mercury ($\sigma = 0.52$ N/m, $\theta = 130°$, $\rho = 13600$ kg/m^3) in the same tube.

Sol: Water: $h = 4\sigma \cos\theta / (\rho g d) = 4 \times 0.0736 \times 1 / (1000 \times 9.81 \times 0.002) = 0.015$ m = 15 mm (rise)

Mercury: $h = 4 \times 0.52 \times \cos 130° / (13600 \times 9.81 \times 0.002) = (2.08 \times -0.6428)/266.8 = -0.00501$ m

→ depression = 5.0 mm

7. Compressibility and Bulk Modulus of Elasticity

Compressibility (β)

Compressibility is the fractional change in volume per unit increase in pressure. It is the reciprocal of bulk modulus. $\beta = 1/K$ (m^2/N or Pa^{-1}).

Bulk Modulus of Elasticity (K)

Ratio of the increase in pressure to the corresponding volumetric strain. It measures resistance of the fluid to compression.

$$K = -dp / (dV / V) = dp / (d\rho / \rho)$$

$$\beta = 1/K$$

- The negative sign shows that volume decreases when pressure increases (dV is negative, so K is positive).
- Unit of K: N/m^2 (Pa). Dimension $M L^{-1} T^{-2}$. Water: $K \approx 2.1 - 2.2$ GPa (2.1×10^9 N/m^2); Mercury ≈ 26 GPa; Oil ≈ 1.5 GPa; Air (at 1 atm, isothermal) = 101 kPa.
- Large K → fluid is nearly incompressible (liquids). Small K → highly compressible (gases).
- For a perfect gas: isothermal process $K = p$; adiabatic process $K = k\,p$ ($k = cp/cv = 1.4$ for air).
- Speed of sound in a fluid: $c = \sqrt{(K/\rho)}$. Water: $c = \sqrt{(2.2 \times 10^9/1000)} \approx 1483$ m/s; air at 15 °C: $c = \sqrt{(kRT)} \approx 340$ m/s.

- Liquids are treated as incompressible in most problems, except water hammer, deep-ocean problems and acoustics.

Example 8

A liquid compressed in a cylinder has a volume of 0.0125 m^3 at 80 N/cm^2 and 0.0124 m^3 at 150 N/cm^2. Find the bulk modulus and compressibility.

Sol: $\Delta p = 150 - 80 = 70\ N/cm^2 = 70 \times 10^4 = 7 \times 10^5\ N/m^2$

$\Delta V = 0.0124 - 0.0125 = -0.0001\ m^3$; $\Delta V/V = -0.0001/0.0125 = -0.008$

$K = -\Delta p/(\Delta V/V) = 7 \times 10^5 / 0.008 = 8.75 \times 10^7\ N/m^2$

$\beta = 1/K = 1.143 \times 10^{-8}\ m^2/N$

Example 9

Water has $K = 2.1 \times 10^9\ N/m^2$. By how much (percent) does its volume decrease when the pressure is raised by 5 MPa?

Sol: $|\Delta V|/V = \Delta p/K = 5 \times 10^6 / 2.1 \times 10^9 = 2.38 \times 10^{-3} = 0.238\ \%$ (so water is practically incompressible)

8. Quick Revision Sheet

Property	Formula	SI Unit
Density	$\rho = m/V$	kg/m^3
Specific weight	$w = \rho g$	N/m^3
Specific gravity	$S = \rho/\rho_{water}$	–
Shear stress	$\tau = \mu\, du/dy$	N/m^2
Dynamic viscosity	$\mu = \tau/(du/dy)$	Pa·s
Kinematic viscosity	$\nu = \mu/\rho$	m^2/s
Droplet pressure	$\Delta p = 4\sigma/d$	N/m^2
Soap bubble	$\Delta p = 8\sigma/d$	N/m^2
Liquid jet	$\Delta p = 2\sigma/d$	N/m^2
Capillary rise	$h = 4\sigma \cos\theta/(\rho g d)$	m
Bulk modulus	$K = -dp/(dV/V)$	N/m^2
Compressibility	$\beta = 1/K$	m^2/N

Important Short Questions

1. Define fluid. How does it differ from a solid?
2. State Newton's law of viscosity. What are Newtonian and non-Newtonian fluids?
3. Why does viscosity of liquids fall and that of gases rise with temperature?
4. Differentiate dynamic and kinematic viscosity; give their units and conversions.
5. What is cavitation? Mention its effects and remedies.
6. Derive the expression for capillary rise in a tube. Why does mercury show depression?
7. Derive the pressure inside a droplet and a soap bubble.
8. Define bulk modulus and compressibility. What is their relation?

Conversion Cheat-Sheet

- 1 poise = 0.1 Pa·s; 1 cP = 10^{-3} Pa·s; 1 stoke = 10^{-4} m^2/s; 1 cSt = 10^{-6} m^2/s
- 1 kgf = 9.81 N; 1 bar = 10^5 Pa; 1 atm = 101.325 kPa = 760 mm Hg; g = 9.81 m/s^2

FLUID MECHANICS

Subject Code: 23A01303

UNIT - 2: Fluid Statics

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Fluid pressure: pressure at a point, Pascal's law, pressure variation with temperature, density and altitude. Piezometer, U-tube manometer, single column manometer, U-tube differential manometer. Pressure gauges. Hydrostatic pressure and force on horizontal, vertical and inclined surfaces. Buoyancy and stability of floating bodies.

1. Fluid Pressure

Pressure

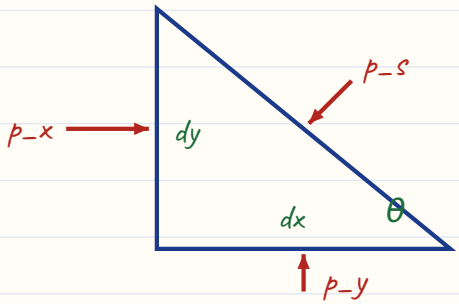
Pressure (intensity of pressure) is the normal force exerted by a fluid per unit area. $p = F/A$. Unit: $N/m^2 = Pa$; $1\ kPa = 10^3\ Pa$; $1\ bar = 10^5\ Pa$.

- Pressure is a scalar; in a fluid at rest it acts normal to any surface (no shear stress when fluid is static).
- $1\ atm = 101.325\ kPa = 760\ mm\ Hg = 10.33\ m\ of\ water = 1.01325\ bar = 14.7\ psi$.

Pressure at a Point – Pascal's Law

Pascal's Law

The pressure at a point in a fluid at rest is the same in all directions: $p_x = p_y = p_z$. Any pressure applied to a confined fluid is transmitted undiminished in all directions.



Element is of unit width

Forces (p × area) in x and y:

$p_x\ dy = p_s\ ds \sin\theta$

$p_y\ dx = p_s\ ds \cos\theta + W/2$

Fig 2.1: Small wedge-shaped element in a static fluid

Proof: Take a small wedge of width unity, base dx, height dy, inclined face ds at angle θ. Being at rest, forces balance in x and y (weight is of second order and neglected).

- x-direction: $p_x\ dy = p_s\ ds \sin\theta = p_s\ dy$ (since $ds \sin\theta = dy$) $\rightarrow p_x = p_s$
- y-direction: $p_y\ dx = p_s\ ds \cos\theta + \tfrac{1}{2} \rho g\ dx\ dy = p_s\ dx$ + (small) $\rightarrow p_y = p_s$ as $dx, dy \rightarrow 0$

- Hence $p_x = p_y = p_s$. The angle θ is arbitrary, so pressure at a point is independent of direction.

Hydraulic Press (Application)



Fig 2.2: Hydraulic lift / press

$$p = F_1/a = W/A \rightarrow W = F_1 \times (A/a)$$

A small force on a small piston creates a large force on a large piston (force multiplication = A/a). Used in hydraulic jacks, brakes, presses and lifts.

Example 1

The diameters of a small and large piston of a hydraulic jack are 5 cm and 25 cm. Find the force needed on the small piston to lift a load of 5000 N.

Sol: $A/a = (25/5)^2 = 25$. $F = W/(A/a) = 5000/25 = 200$ N

2. Hydrostatic Law and Pressure Variation

Consider a vertical fluid column of cross-section dA and height dz in a static fluid (z measured upward). Balancing vertical forces on the element:

$$p\,dA - (p + dp)\,dA - \rho\,g\,dA\,dz = 0 \rightarrow dp/dz = -\rho\,g$$

- This is the basic equation of fluid statics. Pressure increases with depth, and is constant on any horizontal plane in the same continuous fluid.
- Hydrostatic law: The rate of increase of pressure in a vertically downward direction equals the specific weight of the fluid at that point.
- For an incompressible fluid (ρ = const), integrate from the free surface: $p = p_{atm} + \rho\,g\,h$ (absolute), $p_{gauge} = \rho\,g\,h$.
- Pressure head: $h = p/(\rho\,g)$ – the height of a fluid column that produces pressure p. 1 N/cm^2 = 1.02 m of water.

Absolute, Gauge and Vacuum Pressure

Term	Definition	Relation
Absolute pressure	Measured above absolute zero (perfect vacuum)	$p_{abs} = p_{atm} + p_{gauge}$
Gauge pressure	Measured above the local atmospheric pressure	$p_{gauge} = p_{abs} - p_{atm}$
Vacuum (suction) pressure	Pressure below atmospheric, as a positive number	$p_{vac} = p_{atm} - p_{abs}$

Example 2

Find the gauge and absolute pressure at a depth of 10 m in (a) water, (b) oil of $S = 0.8$. Take $p_{atm} = 101.3$ kPa.

Sol: (a) $p_g = \rho g h = 1000 \times 9.81 \times 10 = 98100$ Pa $= 98.1$ kPa; $p_{abs} = 98.1 + 101.3 = 199.4$ kPa

(b) $p_g = 800 \times 9.81 \times 10 = 78480$ Pa $= 78.5$ kPa; $p_{abs} = 78.5 + 101.3 = 179.8$ kPa

Equivalent head of water for oil column $= S \times h = 0.8 \times 10 = 8$ m of water.

Example 2(b) – Pressure in a layered liquid and unit conversion

A tank holds 2 m of oil ($S = 0.8$) floating on 3 m of water. Find the gauge pressure at the bottom. Also express a pressure of 2.5 bar in metres of water and mm of mercury.

Sol: $p = \rho_{oil} g h_1 + \rho_w g h_2 = 800 \times 9.81 \times 2 + 1000 \times 9.81 \times 3 = 15696 + 29430 = 45126$ Pa $= 45.13$ kPa

2.5 bar $= 2.5 \times 10^5$ Pa: $h_w = p/(\rho_w g) = 250000/9810 = 25.48$ m of water; $h_{Hg} = 250000/(13600 \times 9.81) = 1.874$ m $= 1874$ mm Hg

Variation of Pressure with Altitude, Density and Temperature (Compressible Fluids)

For gases the density changes with pressure and temperature. Using the perfect-gas law $p = \rho R T$ in $dp/dz = -\rho g$:

$$\frac{dp}{p} = -\frac{g}{RT}\,dz$$

- Isothermal atmosphere ($T = T_0$ constant): $p = p_0 \exp[-g(z - z_0)/(R T_0)]$ – pressure falls exponentially with altitude.
- Linear temperature variation (troposphere): $T = T_0 - \beta z$, β = lapse rate = 0.0065 K/m up to 11 km. Then
  – $p/p_0 = (1 - \beta z/T_0)^{g/(\beta R)}$ (exponent $g/\beta R = 5.26$ for air)

- Adiabatic (isentropic) atmosphere: $p/\rho^k = \text{const}$, giving $T/T_0 = (p/p_0)^{(k-1)/k}$.
- Density variation: for liquids ρ is almost constant, so p varies linearly with depth. For gases, p and ρ fall with altitude; at constant p, density falls as T rises ($\rho = p/RT$).

Example 3

Find the pressure at an altitude of 3000 m in a standard atmosphere ($T_0 = 288.15$ K, $p_0 = 101.325$ kPa, $\beta = 0.0065$ K/m, $R = 287$ J/kg·K).

Sol: $1 - \beta z/T_0 = 1 - (0.0065 \times 3000)/288.15 = 0.9323$; $g/(\beta R) = 9.81/(0.0065 \times 287) = 5.259$

$p = 101.325 \times (0.9323)^{5.259} = 101.325 \times 0.6917 = 70.1$ kPa

3. Measurement of Pressure – Manometers

Manometers are devices that measure pressure by balancing it against a column of liquid. Gauge pressure is found by using the hydrostatic law.

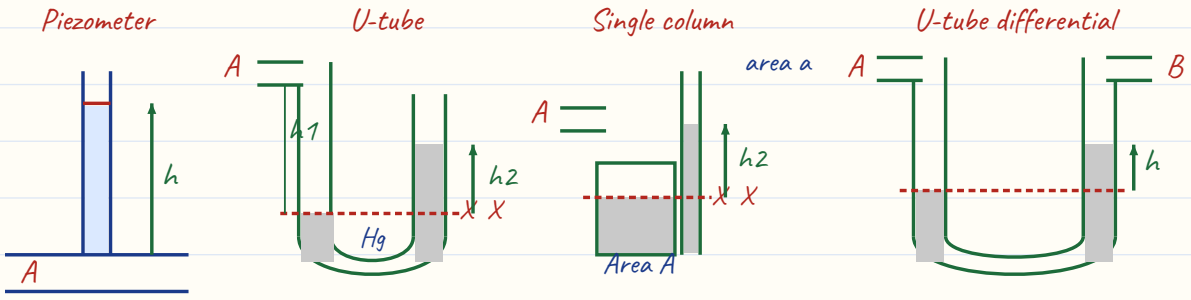


Fig 2.3: Types of manometers

Rules for Solving Manometer Problems

1. Choose a datum line (X–X) through the lowest common surface of the same liquid.
2. Pressures at the same level in the same continuous liquid are equal.
3. Moving down in a liquid → add $\rho g h$; moving up → subtract $\rho g h$.
4. Start from one end and write the pressure up to the other end (open to atmosphere → gauge pressure zero).

(a) Piezometer

The simplest manometer: a vertical glass tube open at the top, fixed to the point where pressure is required.

$$p_A = \rho g h$$

- Only for moderate gauge pressures of liquids; cannot measure gas pressure (gas escapes) or large pressures (tube too tall), nor negative pressure.

- Tube bore should be > 12 mm to avoid capillary error.

(b) U-Tube Manometer

A U-shaped glass tube holds a heavy manometric liquid (mercury, $S = 13.6$) which is immiscible with the fluid whose pressure is to be measured; one end connected to the point, the other open to atmosphere.

$$p_A + \rho_1 g h_1 = \rho_2 g h_2 \rightarrow p_A = \rho_2 g h_2 - \rho_1 g h_1$$

- ρ_1 = density of the fluid in pipe, ρ_2 = density of the manometric liquid, h_1 = height of pipe fluid above X–X, h_2 = height of manometric liquid above X–X in the open limb.
- For a gas in pipe, $\rho_1 g h_1$ is negligible: $p_A = \rho_2 g h_2$. For vacuum (negative) pressure the mercury is pulled up on the pipe side and $p_A = -\rho_2 g h_2 - \rho_1 g h_1$.

Example 4

A U-tube manometer connected to a pipe containing oil of $S = 0.9$ shows mercury ($S = 13.6$) 200 mm above the datum in the open limb. The oil column above the datum in the pipe limb is 100 mm. Find the gauge pressure in the pipe.

Sol: $p_A + \rho_{oil} g (0.1) = \rho_{Hg} g (0.2)$

$p_A = 13600 \times 9.81 \times 0.2 - 900 \times 9.81 \times 0.1 = 26683 - 883 = 25800\ N/m^2 = 25.8\ kPa$ (gauge)

(c) Single Column Manometer (Micromanometer type)

A modification of the U-tube with a large reservoir (area A) on one side and a narrow tube (area a) on the other. Because $A \gg a$, the level fall in the reservoir is very small, so only the rise h_2 in the narrow limb is read – more sensitive.

When pressure p_A is applied to the reservoir, mercury falls Δh in the reservoir and rises h_2 in the narrow limb. Volume balance: $A \Delta h = a h_2 \rightarrow \Delta h = (a/A) h_2$.

$$p_A + \rho_1 g (h_1 + \Delta h) = \rho_2 g (h_2 + \Delta h)$$

$$p_A = \rho_2 g h_2 (1 + a/A) - \rho_1 g h_1 (1 + \ldots) \approx \rho_2 g (h_2 + \Delta h) - \rho_1 g (h_1 + \Delta h)$$

- If a/A is very small: $p_A \approx \rho_2 g h_2 - \rho_1 g h_1$. The inclined-tube version further magnifies the reading ($h_2 = l \sin\theta$).

Example 5

A single column manometer is connected to a water pipe. Reservoir/tube area ratio A/a = 100; h_1 = 0.2 m (water above datum), h_2 = 0.5 m (mercury rise in the narrow limb). Find p_A.

Sol: $\Delta h = (a/A)\, h_2 = 0.5/100 = 0.005$ m

$p_A = \rho_{Hg}\, g\, (h_2 + \Delta h) - \rho_w\, g\, (h_1 + \Delta h) = 13600 \times 9.81 \times 0.505 - 1000 \times 9.81 \times 0.205$

$= 67375 - 2011 = 65364\ N/m^2 = 65.4$ kPa

(d) U-Tube Differential Manometer

Measures the difference of pressure between two points A and B in the same pipe or in two different pipes. Both ends are connected to the points; the U-tube holds a heavy liquid.

$$p_A + \rho_1 g\,(h + y) = p_B + \rho_1 g\, y + \rho_m g\, h$$

$$p_A - p_B = g\, h\, (\rho_m - \rho_1)$$ (A, B at same level, same fluid ρ_1)

- If B is lower than A or the fluids differ, write the equation separately for each limb (add/subtract heads as per the rules).
- Inverted U-tube differential manometer uses a light liquid (air or oil) on top for measuring small pressure differences in liquids: $p_A - p_B = \rho_1 g\, h_1 - \rho_s g\, h_s - \rho_2 g\, h_2$ (heads in respective legs).

Example 6

Two pipes A and B at the same level carry water. A differential U-tube manometer containing mercury shows a deflection of 0.3 m. Find $p_A - p_B$.

Sol: $p_A - p_B = g\, h\, (\rho_m - \rho_w) = 9.81 \times 0.3 \times (13600 - 1000) = 9.81 \times 0.3 \times 12600$

$= 37082\ N/m^2 = 37.08$ kPa (head difference = 3.78 m of water)

Mechanical Pressure Gauges

Manometers are not suitable for very high pressures or fluctuating pressures, so mechanical gauges are used.

- Bourdon tube gauge: a curved, flattened elliptical metal tube closed at one end. Pressure inside tends to straighten the tube; the small movement is magnified through a link, sector and pinion to a pointer on a dial. Measures gauge pressure up to ~1000 bar.
- Diaphragm gauge: pressure deflects a thin flexible diaphragm; used for low pressures and corrosive fluids.
- Dead-weight pressure gauge: piston loaded with known weights; used to calibrate other gauges ($p = W/A$).

- Bellows gauge – low pressures; Vacuum gauge – measures pressures below atmosphere; Pressure transducers – electrical (strain gauge, piezoelectric).

4. Hydrostatic Forces on Submerged Surfaces

A fluid at rest exerts a normal force on any submerged surface. We need (i) the total pressure (magnitude of force) and (ii) the centre of pressure (point of application).

(a) Horizontal Plane Surface

Pressure is uniform over the surface (all points at the same depth h):

$$F = p\,A = \rho\,g\,h\,A \text{ acts at the centroid (centre of pressure = centroid)}$$

(b) Vertical and Inclined Plane Surfaces

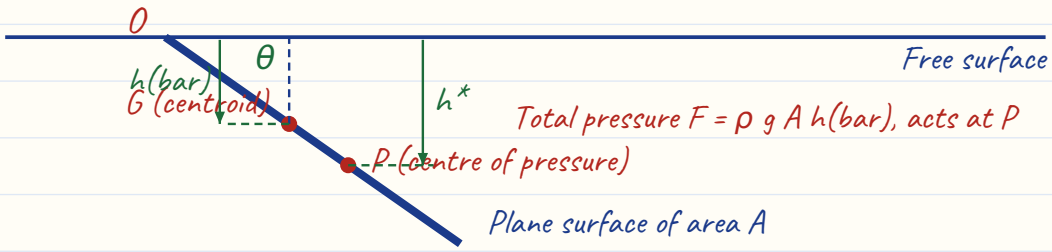


Fig 2.4: Inclined plane surface submerged in a liquid

Let the plane be inclined at angle θ to the free surface, A = area, h(bar) = depth of centroid G below the free surface; y = distance measured along the plane from O (the intersection with free surface). Pressure on a strip dA at depth $h = y\sin\theta$ is $\rho\,g\,y\sin\theta$.

- Total pressure: $F = \int \rho\,g\,y\sin\theta\,dA = \rho\,g\sin\theta\int y\,dA = \rho\,g\sin\theta\cdot A\,y(bar) = \rho\,g\,A\,h(bar)$
- Hence total force = (specific weight) × (area) × (depth of centroid). It is independent of the angle θ for a given centroid depth.

Centre of pressure (h*): Take moments of the elemental forces about O:

$$F\cdot y^* = \int \rho\,g\,y^2\sin\theta\,dA = \rho\,g\sin\theta\cdot I_O$$

using parallel axis theorem $I_O = I_G + A\,y_c^2 \rightarrow$

$$h^* = h(bar) + (I_G\sin^2\theta)\,/\,(A\,h(bar))$$

- I_G = second moment of area about the horizontal axis through the centroid. For a vertical surface (θ = 90°): $h^* = h(bar) + I_G/(A\,h(bar))$.
- The centre of pressure is always below the centroid ($h^* > h(bar)$) because pressure increases with depth; it approaches G as the depth increases.

- Horizontal plane: $\sin\theta = 0$, so $h^* = h(\text{bar})$ (coincide).

Shape	Area A	I_G (about horizontal axis through centroid)	Centroid
Rectangle (b wide, d deep)	$b\,d$	$b\,d^3/12$	$d/2$ from top
Triangle (base b, height d)	$\frac{1}{2}\,b\,d$	$b\,d^3/36$	$d/3$ from base
Circle (dia d)	$\pi\,d^2/4$	$\pi\,d^4/64$	centre
Semicircle (radius r)	$\pi\,r^2/2$	$0.1098\,r^4$	$4r/3\pi$ from base

Example 7 – Vertical rectangular gate

A rectangular gate 2 m wide and 3 m high is vertical in a water tank, its top edge at the free surface. Find the total force and depth of centre of pressure.

Sol: $A = 2 \times 3 = 6\ m^2$; $h(\text{bar}) = 1.5\ m$

$F = \rho\,g\,A\,h(\text{bar}) = 9810 \times 6 \times 1.5 = 88290\ N = 88.3\ kN$

$I_G = b\,d^3/12 = 2 \times 27/12 = 4.5\ m^4$; $h^* = h(\text{bar}) + I_G/(A\,h(\text{bar})) = 1.5 + 4.5/(6 \times 1.5) = 2.0\ m$ (= 2/3 of depth, from the free surface)

Example 8 – Inclined circular plate

A circular plate of diameter 2 m is inclined at 30° to the horizontal. Its centre is 4 m below the free surface of water. Find F and the centre of pressure.

Sol: $A = \pi(2)^2/4 = 3.1416\ m^2$; $h(\text{bar}) = 4\ m$

$F = \rho\,g\,A\,h(\text{bar}) = 9810 \times 3.1416 \times 4 = 123276\ N = 123.3\ kN$

$I_G = \pi\,d^4/64 = \pi \times 16/64 = 0.7854\ m^4$; $h^* = 4 + (0.7854 \times \sin^2 30°)/(3.1416 \times 4) = 4 + 0.7854 \times 0.25/12.566 = 4.0156\ m$

Distance along the plate from the free surface $= h^*/\sin\theta = 4.0156/0.5 = 8.03\ m$

Example 8(b) – Vertical triangular plate

A triangular plate of base 3 m and height 2 m is submerged vertically in water with its base horizontal at the free surface and apex downward. Find the force and centre of pressure.

Sol: $A = \frac{1}{2} \times 3 \times 2 = 3\ m^2$; centroid is $d/3 = 0.667\ m$ below the base (free surface): $h(\text{bar}) = 0.667\ m$

$F = \rho\,g\,A\,h(\text{bar}) = 9810 \times 3 \times 0.667 = 19620\ N = 19.62\ kN$

$I_G = b\,d^3/36 = 3 \times 8/36 = 0.667\ m^4$; $h^* = 0.667 + 0.667/(3 \times 0.667) = 1.0\ m$ below the free surface

Remarks

- Pressure diagram (prism): For a rectangular vertical surface, the load is a triangular prism; F acts at its centroid, 2/3 depth from the top.
- If there is liquid on both sides, the net force is the difference of forces and the centre of pressure is found by moments.
- For a curved surface, find horizontal component F_H = force on vertical projection and vertical component F_V = weight of liquid above the surface; resultant $F = \sqrt{(F_H^2 + F_V^2)}$.
- Applications: design of dams, sluice gates, tank walls, lock gates.

5. Buoyancy and Stability of Floating Bodies

Archimedes' Principle
A body wholly or partly immersed in a fluid experiences an upward force (buoyant force) equal to the weight of the fluid displaced by the body.

$$F_B = \rho_{fluid}\, g\, V_{displaced}$$

- Centre of buoyancy (B): the centroid of the displaced volume of fluid; the buoyant force acts through B.
- Proof: upward force on the bottom of a prism = $\rho\, g\, (h + z)\, A$; downward on top = $\rho\, g\, h\, A$; net = $\rho\, g\, z\, A$ = weight of displaced fluid.
- Conditions: $W > F_B$ → body sinks; $W = F_B$ → floats (neutrally buoyant if fully submerged); $W < F_B$ → rises until it floats, with $F_B = W$.
- Law of floatation: A floating body displaces its own weight of fluid; $\rho_b\, V = \rho_f\, V_d$
  Fraction submerged = S_b / S_f
- Applications: ships, submarines, hydrometers (density), balloons, life jackets.

Stability of Submerged and Floating Bodies

- Submerged body – the weight acts at the centre of gravity G and the buoyancy at B. Stable if B is above G (restoring couple on tilting); unstable if B is below G; neutral if B and G coincide.
- Floating body – on tilting, the shape of the displaced volume changes, so B moves to a new position B'. The vertical line through B' cuts the original axis at the metacentre M.

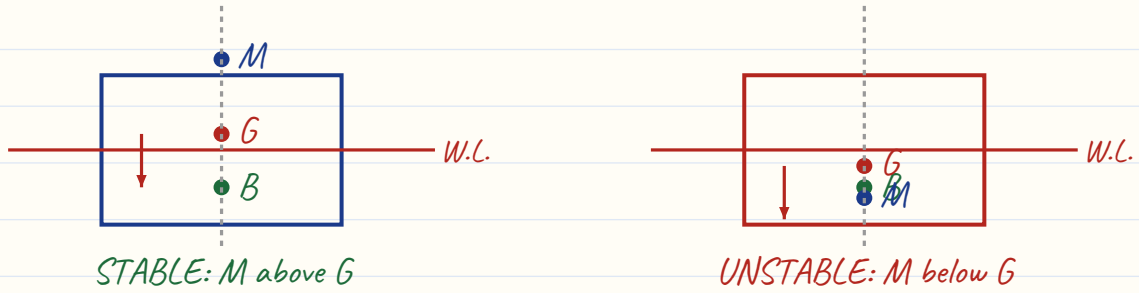


Fig 2.5: Equilibrium conditions of a floating body (B = centre of buoyancy, G = centre of gravity, M = metacentre)

Metacentre (M) and Metacentric Height (GM)

The metacentre is the point about which a floating body starts oscillating when given a small angular displacement; it is the intersection of the line of action of buoyancy (in the displaced position) with the original vertical axis. Metacentric height GM is the distance between G and M.

$$BM = I / V$$

$$GM = BM - BG = I/V - BG$$

- I = second moment of area of the waterline (floating) section about the longitudinal axis (axis of tilt); V = volume of fluid displaced; BG = distance between B and G (positive if G above B).
- Stable equilibrium: M above G ($GM > 0$) – the couple $W \times GM \sin\theta$ restores the body.
- Unstable: M below G ($GM < 0$) – the couple overturns the body. Neutral: M coincides with G ($GM = 0$).
- Righting moment = $W \cdot GM \cdot \sin\theta \approx W \cdot GM \cdot \theta$ (for small angles).
- Time period of rolling: $T = 2\pi \sqrt{(k^2 / (g \cdot GM))}$, k = radius of gyration about the axis of roll.

Experimental Determination of Metacentric Height

A pontoon of weight W floats in water. A movable weight w is placed at the centre and then shifted sideways by a distance x; the angle of heel θ is measured with a plumb-line.

$$GM = (w \cdot x) / (W \cdot \tan\theta)$$ (W = total weight including w)

Example 9 – Metacentric height of a rectangular block

A rectangular block 1.5 m wide, 0.8 m deep and 4 m long has $S = 0.7$ and floats in water with its 4 m side horizontal along the axis. Check its stability for rolling about the longitudinal axis.

Sol: Draught $d = S \times D = 0.7 \times 0.8 = 0.56$ m. G above base $= D/2 = 0.4$ m; B above base $= d/2 = 0.28$ m; $BG = 0.12$ m

$I = L B^3/12 = 4 \times 1.5^3/12 = 1.125\ m^4$; $V = 4 \times 1.5 \times 0.56 = 3.36\ m^3$

$BM = I/V = 0.3348$ m; $GM = BM - BG = 0.3348 - 0.12 = 0.215$ m

GM positive → M above G → the block is stable.

Example 10 – Buoyancy

A wooden block of $S = 0.6$ floats in water. What percentage of its volume is submerged? If a body of 0.5 m^3 is fully immersed in water, what is the buoyant force?

Sol: Fraction submerged $= S_b/S_f = 0.6 \rightarrow 60\ \%$. $F_B = \rho g V = 1000 \times 9.81 \times 0.5 = 4905\ N$

Example 11 – Density of a stone by buoyancy

A stone weighs 392.4 N in air and 250 N when fully immersed in water. Find its volume, density and specific gravity.

Sol: Buoyant force $= 392.4 - 250 = 142.4\ N = \rho_w g V \rightarrow V = 142.4/9810 = 0.01452\ m^3$

Mass $= 392.4/9.81 = 40$ kg; $\rho = 40/0.01452 = 2756\ kg/m^3$; $S = 2.76$

Stability – Points to Remember

- For wide, shallow vessels (large waterline I) BM is large → good stability (barges, pontoons). Narrow, deep bodies tend to roll over.
- Stability can be improved by lowering G (ballast at bottom) or increasing the beam (width) of the vessel.
- A cylinder or sphere floating with its axis vertical must be checked for stability: for a floating sphere, GM depends on the density ratio.
- Ship designers use the metacentric height GM of about 0.3 – 1.5 m; too large a GM makes rolling quick and uncomfortable (stiff ship), too small makes the ship tender and dangerous.

6. Quick Revision Sheet

Concept	Formula
Pascal's law	$p_x = p_y = p_z$; $W = F(A/a)$
Hydrostatic law	$dp/dz = -\rho g$; $p = \rho g h$
Absolute pressure	$p_{abs} = p_{atm} + p_{gauge}$
U-tube manometer	$p_A = \rho_2 g h_2 - \rho_1 g h_1$
Differential manometer	$p_A - p_B = g h (\rho_m - \rho_1)$
Total pressure on plane	$F = \rho g A h(bar)$
Centre of pressure	$h^* = h(bar) + I_G \sin^2\theta / (A h(bar))$
Buoyancy	$F_B = \rho g V_d$
Metacentric height	$GM = I/V - BG$ (stable if $GM > 0$)

Important Questions

1. State and prove Pascal's law.
2. Derive the hydrostatic law and obtain pressure variation in an isothermal and a linear-temperature atmosphere.
3. Compare absolute, gauge and vacuum pressure; explain the working of a Bourdon tube gauge.
4. Explain piezometer, U-tube manometer, single-column manometer and differential manometer with derivations.
5. Derive expressions for total pressure and centre of pressure on an inclined plane surface.
6. Explain buoyancy, centre of buoyancy, metacentre and conditions of stability of a floating body.

FLUID MECHANICS

Subject Code: 23A01303

UNIT - 3: Fluid Kinematics

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Classification of fluid flow: steady and unsteady; uniform and non-uniform; laminar and turbulent; rotational and irrotational; compressible and incompressible; ideal and real; one, two and three dimensional flows. Stream line, path line, streak line and stream tube; stream function, velocity potential function. One, two and three-dimensional continuity equations in Cartesian coordinates.

1. Introduction to Fluid Kinematics

Fluid Kinematics

Kinematics is the study of fluid motion in terms of displacement, velocity and acceleration, without considering the forces or energy that cause the motion.

Methods of Describing Fluid Motion

- Lagrangian method: follows a single fluid particle through space and time (position $x(t)$, $y(t)$, $z(t)$). Like tracking one car on a road. Difficult for fluids because of huge number of particles.
- Eulerian method: observes the flow at fixed points in space as time passes. Velocity is a function of position and time: $V = V(x, y, z, t)$. Like a traffic counter at a junction. Most commonly used in fluid mechanics.

$$V = u\,i + v\,j + w\,k \text{ with } u = u(x, y, z, t),\ v = v(x, y, z, t),\ w = w(x, y, z, t)$$

Acceleration of a Fluid Particle (Material Derivative)

Using the chain rule, the acceleration components of a fluid particle are:

$$a_x = \partial u/\partial t + u\,\partial u/\partial x + v\,\partial u/\partial y + w\,\partial u/\partial z$$
$$a_y = \partial v/\partial t + u\,\partial v/\partial x + v\,\partial v/\partial y + w\,\partial v/\partial z$$
$$a_z = \partial w/\partial t + u\,\partial w/\partial x + v\,\partial w/\partial y + w\,\partial w/\partial z$$

- Local (temporal) acceleration = $\partial u/\partial t$ etc. – change of velocity with time at a fixed point. It is zero in steady flow.

- Convective acceleration = $u\,\partial u/\partial x + v\,\partial u/\partial y + w\,\partial u/\partial z$ – change of velocity due to change of position. It is zero in uniform flow.
- Total acceleration = local + convective = DV/Dt (substantial or material derivative).

Example 1

For a steady 2-D flow $u = x^2y$, $v = -xy^2$, find the acceleration at the point (1, 2).

Sol: Steady → $\partial u/\partial t = \partial v/\partial t = 0$. $\partial u/\partial x = 2xy$, $\partial u/\partial y = x^2$, $\partial v/\partial x = -y^2$, $\partial v/\partial y = -2xy$

$a_x = u\,\partial u/\partial x + v\,\partial u/\partial y = x^2y(2xy) + (-xy^2)(x^2) = x^3y^2$

$a_y = u\,\partial v/\partial x + v\,\partial v/\partial y = x^2y(-y^2) + (-xy^2)(-2xy) = x^2y^3$

At (1, 2): $a_x = 1 \times 4 = 4\ m/s^2$, $a_y = 1 \times 8 = 8\ m/s^2$, $|a| = \sqrt{(16 + 64)} = 8.94\ m/s^2$

Example 1(b) – Acceleration in a tapering pipe

A pipe tapers linearly from $d = 0.3$ m at $x = 0$ to $d = 0.15$ m at $x = 3$ m. Steady discharge $Q = 0.1\ m^3/s$. Find the convective acceleration at both ends. If Q rises at $0.02\ m^3/s$ per second, find the total acceleration at the inlet.

Sol: $d = 0.3 - 0.05x$; $V = 4Q/(\pi d^2)$; $dV/dx = (-8Q/\pi d^3)(dd/dx) = 0.4Q/(\pi d^3)$

At $x = 0$: $V = 1.415\ m/s$; $dV/dx = 0.4716\ s^{-1}$; $a_{conv} = V\,dV/dx = 0.667\ m/s^2$

At $x = 3$: $V = 5.659\ m/s$; $dV/dx = 3.772\ s^{-1}$; $a_{conv} = 21.35\ m/s^2$

Local acceleration at inlet = $(4/\pi d^2)\,dQ/dt = 4 \times 0.02/(\pi \times 0.09) = 0.283\ m/s^2$; total = $0.667 + 0.283 = 0.95\ m/s^2$

Lagrangian method	Eulerian method
Follows an individual fluid particle	Observes fixed points in space
Variables: position of particle vs time	Variables: velocity field $V(x, y, z, t)$
Mathematically complex for fluids	Simpler; used in most of fluid mechanics (and in CFD)
Gives pathlines directly	Gives streamlines directly

2. Classification of Fluid Flow

(a) Steady and Unsteady Flow

- Steady flow: fluid properties (velocity, pressure, density) at a point do not change with time. $\partial V/\partial t = 0$, $\partial p/\partial t = 0$, $\partial \rho/\partial t = 0$. Eg: flow through a pipe at a constant discharge.
- Unsteady flow: properties at a point change with time: $\partial V/\partial t \neq 0$. Eg: flow in a pipe while a valve is opening, rising/falling river during a flood, a tank being emptied.

(b) Uniform and Non-uniform Flow

- Uniform flow: velocity is the same in magnitude and direction at every point at a given instant: $\partial V/\partial s = 0$. Eg: flow through a long pipe of constant diameter (steady), liquid under constant pressure in a straight constant-section pipe.
- Non-uniform flow: velocity changes from point to point at a given instant: $\partial V/\partial s \neq 0$. Eg: flow through a converging/diverging pipe or around a bend.

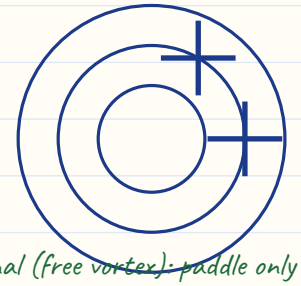
Note: Four combinations are possible: steady uniform (constant-diameter pipe, constant Q); steady non-uniform (tapering pipe, constant Q); unsteady uniform (constant-diameter pipe with varying Q); unsteady non-uniform (tapering pipe with varying Q).

(c) Laminar and Turbulent Flow

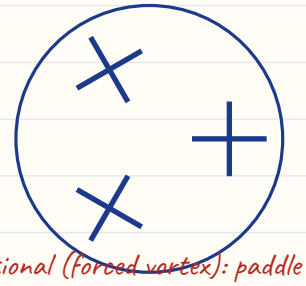
- Laminar flow: fluid particles move in smooth, parallel layers (streamlines) with no mixing; viscous forces dominate. Occurs at low velocity / high viscosity. Reynolds number $Re < 2000$ (in pipes).
- Turbulent flow: random, irregular motion with eddies, mixing and velocity fluctuations; inertial forces dominate. $Re > 4000$ (in pipes).
- Transitional flow: $2000 < Re < 4000$.
- Reynolds number $Re = \rho V D / \mu = V D / \nu$ (ratio of inertia force to viscous force).

(d) Rotational and Irrotational Flow

- Rotational flow: fluid particles rotate about their own mass centre as they move. Eg: flow near a solid boundary (boundary layer), forced vortex.
- Irrotational flow: no rotation of particles about their own axes (they may move in a circular path but the orientation is unchanged). Eg: free vortex (outside core), flow of ideal fluid, flow far from boundaries.
- Test: rotation components $\omega_z = \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y)$, $\omega_x = \tfrac{1}{2}(\partial w/\partial y - \partial v/\partial z)$, $\omega_y = \tfrac{1}{2}(\partial u/\partial z - \partial w/\partial x)$. Irrotational if $\omega_x = \omega_y = \omega_z = 0$.
- Vorticity $= 2\omega$ (twice the rotation); circulation $\Gamma = \oint V \cdot ds = \iint$ (vorticity) dA (Stokes theorem).



Irrotational (free vortex): paddle only translates



Rotational (forced vortex): paddle spins

Fig 3.1: Rotational and irrotational flow

(e) Compressible and Incompressible Flow

- Incompressible flow: density is constant throughout the flow field ($\partial\rho/\partial t = 0$ and $\nabla\rho = 0$). All liquids; gases at Mach number < 0.3.
- Compressible flow: density varies from point to point; $\rho \neq$ const. Gases at high speeds (Mach number > 0.3), e.g. high-speed gas flow in nozzles, aircraft aerodynamics, shock waves.

(f) Ideal and Real Fluid Flow

- Ideal fluid flow: fluid has zero viscosity (frictionless) and is incompressible. No shear stress, no boundary layer; potential flow theory applies. Imaginary but a good approximation away from solid walls.
- Real fluid flow: viscosity is present, giving shear stresses, friction losses, boundary layers and no-slip at walls.

(g) One, Two and Three Dimensional Flow

- One-dimensional flow: flow parameters (velocity) are functions of one space coordinate (and time) only: $V = f(x)$. Eg: flow through a pipe (average velocity along the axis).
- Two-dimensional flow: velocity is a function of two coordinates; velocity in the third direction is zero or negligible: $u = f(x, y)$, $v = f(x, y)$, $w = 0$. Eg: flow over a very long dam/wing section, flow between wide parallel plates.
- Three-dimensional flow: velocity components are functions of all three coordinates: $u, v, w = f(x, y, z)$. The most general case. Eg: flow around a ship, flow in a river bend, near a pipe junction.

Classification	Criterion	Example
Steady / unsteady	$\partial(\ )/\partial t = 0$ or $\neq 0$	Constant-head tank discharge / emptying tank
Uniform / non-uniform	$\partial V/\partial s = 0$ or $\neq 0$	Constant-section pipe / tapering pipe
Laminar / turbulent	$Re < 2000$ / $Re > 4000$	Oil in a thin tube / water in a main
Rotational / irrotational	$\omega \neq 0$ / $\omega = 0$	Near wall / free vortex
Compressible / incompressible	ρ varies / ρ = const	High-speed gas / water
Ideal / real	$\mu = 0$ / $\mu \neq 0$	Potential flow / actual flows
1-D, 2-D, 3-D	No. of space coordinates	Pipe / wide channel / ship hull

3. Flow Lines – Streamline, Pathline, Streakline, Stream Tube

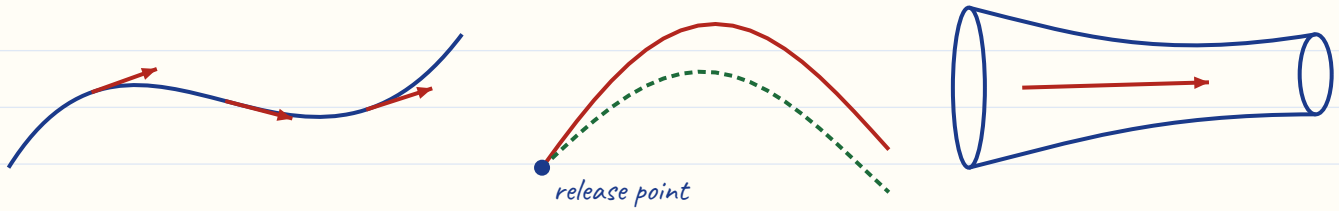


Fig 3.2: Flow lines (streamline, path/streak line and stream tube)

Streamline
An imaginary curve drawn in the flow field such that the tangent at every point gives the direction of velocity vector at that point at that instant.

- Differential equation of a streamline: $dx/u = dy/v = dz/w$.
- Streamlines cannot intersect (otherwise a particle would have two velocities at one point) and there is no flow across a streamline.
- In steady flow the streamline pattern does not change with time; in unsteady flow it changes.

Pathline
The actual path traced by a single fluid particle over a period of time (Lagrangian concept, like a long-exposure photograph of one particle).

Streakline
The locus of all fluid particles that have passed through a fixed point in the flow field (like the smoke trail from a chimney or dye injected continuously at a point).

Stream Tube
An imaginary tube formed by a bundle of streamlines passing through a closed curve. No fluid crosses the walls of the tube; fluid enters and leaves only at the ends. It behaves like a pipe with impermeable walls.

Note: In steady flow, the streamline, pathline and streakline coincide. In unsteady flow they are different curves.

Example 2 – Equation of a streamline

Find the streamline through (1, 1) for the 2-D flow $u = x$, $v = -y$.

Sol: $dx/u = dy/v \rightarrow dx/x = -dy/y \rightarrow \ln x = -\ln y + \ln C \rightarrow xy = C$. At (1, 1): $C = 1 \rightarrow$ streamline is the rectangular hyperbola $xy = 1$.

4. Rate of Flow and Continuity Equation

Discharge, Mean Velocity

$$Q = \int V \, dA = A V_{mean} \quad (m^3/s)$$

$$\text{Mass flow rate } \dot{m} = \rho A V \quad (kg/s)$$

(a) Continuity Equation – One-Dimensional Flow (Law of Conservation of Mass)



Fig 3.3: Flow through a converging pipe

For steady flow through a stream tube, the mass entering section 1 per unit time equals the mass leaving section 2 (no accumulation, no sources/sinks):

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad \text{(compressible, steady)}$$

$$A_1 V_1 = A_2 V_2 = Q = \text{constant} \quad \text{(incompressible)}$$

- Since area decreases, velocity increases in a converging pipe; $V \propto 1/A$ ($V \propto 1/d^2$).
- For a pipe junction (branching): $Q_1 = Q_2 + Q_3$ (inflow = outflow).
- For unsteady one-dimensional flow: $\partial(\rho A)/\partial t + \partial(\rho A V)/\partial x = 0$.

Example 3 – Converging pipe

Water flows at 0.05 m^3/s through a pipe that reduces from 300 mm to 200 mm diameter. Find the velocities at the two sections.

Sol: $A_1 = \pi(0.3)^2/4 = 0.07069\ m^2$; $V_1 = Q/A_1 = 0.05/0.07069 = 0.707\ m/s$

$A_2 = \pi(0.2)^2/4 = 0.03142\ m^2$; $V_2 = 0.05/0.03142 = 1.592\ m/s$ (check: $V_2 = V_1 (d_1/d_2)^2 = 0.707 \times 2.25$ ✓)

Example 4 – Branching pipe

A 600 mm main pipe carrying water at 3 m/s branches into 300 mm and 400 mm pipes. If the velocity in the 300 mm pipe is 2 m/s, find the discharge in the main pipe and the velocity in the 400 mm pipe.

Sol: $Q = A\,V = (\pi \times 0.36/4) \times 3 = 0.2827 \times 3 = 0.848\ m^3/s$

$Q_1 = (\pi \times 0.09/4) \times 2 = 0.07069 \times 2 = 0.1414\ m^3/s$; $Q_2 = Q - Q_1 = 0.848 - 0.141 = 0.7068\ m^3/s$

$V_2 = Q_2/A_2 = 0.7068/(\pi \times 0.16/4 = 0.12566) = 5.63\ m/s$

Example 5 – Compressible (gas) flow

Air flows in a duct: at section 1 d = 0.3 m, V = 20 m/s, ρ = 1.2 kg/m^3. At section 2 d = 0.2 m and ρ = 1.0 kg/m^3. Find V_2.

Sol: $\dot{m} = \rho_1 A_1 V_1 = 1.2 \times 0.07069 \times 20 = 1.697\ kg/s$; $V_2 = \dot{m}/(\rho_2 A_2) = 1.697/(1.0 \times 0.03142) = 54.0\ m/s$

(b) Continuity Equation – Three-Dimensional Flow in Cartesian Coordinates

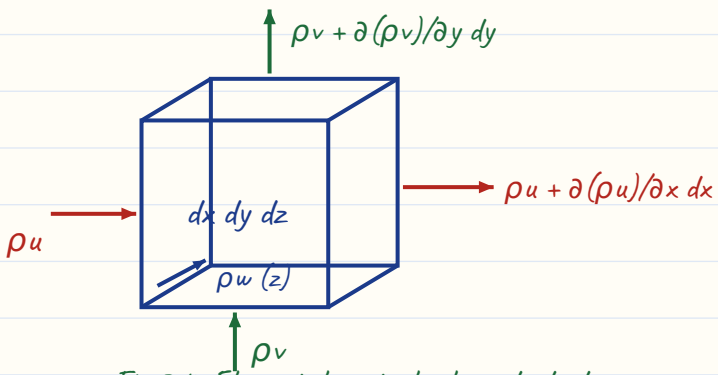


Fig 3.4: Elemental control volume dx dy dz

Consider a fixed rectangular control volume dx · dy · dz in a flow with velocity components u, v, w. The mass entering through face perpendicular to x-axis is ρ u dy dz, and the mass leaving is [ρu + ∂(ρu)/∂x dx] dy dz.

- Net mass outflow along x = ∂(ρu)/∂x · dx dy dz
- Net mass outflow along y = ∂(ρv)/∂y · dx dy dz; along z = ∂(ρw)/∂z · dx dy dz

- Rate of mass accumulation in the element = $\partial\rho/\partial t \cdot dx\,dy\,dz$
- Net outflow + rate of accumulation = 0 (mass is conserved).

$$\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$$

(Unsteady compressible 3-D continuity equation)

Special Cases

Case	Continuity equation
Steady, compressible, 3-D ($\partial\rho/\partial t = 0$)	$\partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$
Incompressible, 3-D (ρ = const)	$\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$ ($\nabla \cdot V = 0$)
Incompressible, 2-D ($w = 0$)	$\partial u/\partial x + \partial v/\partial y = 0$
Incompressible, 1-D	$\partial u/\partial x = 0 \rightarrow u$ = constant (A = const)
Vector form (compressible)	$\partial\rho/\partial t + \nabla \cdot (\rho V) = 0$

Note: In cylindrical polar coordinates (r, θ, z) for incompressible flow: $\partial(r v_r)/(r\,\partial r) + (1/r)\,\partial v_\theta/\partial\theta + \partial v_z/\partial z = 0$.

Example 6 – Check of continuity

Which of the following flows is possible for an incompressible fluid? (i) $u = x^2 + y^2$, $v = -2xy$ (ii) $u = 4xy + y^2$, $v = 6xy + 3x$

Sol: (i) $\partial u/\partial x = 2x$, $\partial v/\partial y = -2x \rightarrow$ sum = 0 ✓ → possible

(ii) $\partial u/\partial x = 4y$, $\partial v/\partial y = 6x \rightarrow$ sum = $4y + 6x \neq 0$ ✗ → not possible

Example 7 – Finding the missing velocity component

For steady incompressible flow, $u = 4x$ and $v = 0$ at $y = 0$. Find v for 2-D flow. For a 3-D flow with $u = 2x$, $v = 3y$ find w, given $w = 0$ at $z = 0$.

Sol: 2-D: $\partial v/\partial y = -\partial u/\partial x = -4 \rightarrow v = -4y + f(x)$; at $y = 0$ $v = 0 \rightarrow v = -4y$

3-D: $\partial w/\partial z = -(2 + 3) = -5 \rightarrow w = -5z + f(x,y)$; $w = 0$ at $z = 0 \rightarrow w = -5z$

5. Rotation, Vorticity and Circulation

The rotation of a fluid particle about the z-axis is the average angular velocity of two mutually perpendicular lines:

$$\omega_z = \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y),\ \text{Vorticity}\ \zeta = 2\omega_z = \partial v/\partial x - \partial u/\partial y$$

- Flow is irrotational if $\partial v/\partial x = \partial u/\partial y$ (for 2-D).

- Circulation $\Gamma = \oint V \cos\alpha \, ds$ around a closed curve $= \iint \zeta \, dA$ (Stokes' theorem). For irrotational flow $\Gamma = 0$ around any closed path not enclosing a singularity (a vortex core).
- Angular deformation (shear strain rate) $= \partial v/\partial x + \partial u/\partial y$; linear deformation $= \partial u/\partial x$, $\partial v/\partial y$.

Example 8

Check whether the flows are rotational: (i) $u = y$, $v = 0$ (simple shear); (ii) $u = -\Omega y$, $v = \Omega x$ (solid-body rotation); (iii) $u = -y/(x^2+y^2)$, $v = x/(x^2+y^2)$ (free vortex).

Sol: (i) $\omega_z = \frac{1}{2}(0 - 1) = -0.5 \rightarrow$ rotational. (ii) $\omega_z = \frac{1}{2}(\Omega + \Omega) = \Omega \rightarrow$ rotational (every particle spins with Ω).

(iii) $\partial v/\partial x = (y^2 - x^2)/(x^2+y^2)^2$ and $\partial u/\partial y = (y^2 - x^2)/(x^2+y^2)^2 \rightarrow$ equal $\rightarrow \omega_z = 0 \rightarrow$ irrotational (except at the origin).

6. Stream Function (ψ)

Stream function

A scalar function of space and time, $\psi(x, y, t)$, defined only for two-dimensional incompressible flow, whose partial derivative with respect to any direction gives the velocity component at right angles to that direction.

$$u = \partial\psi/\partial y \, , \; v = -\partial\psi/\partial x$$

- Substitution into continuity: $\partial u/\partial x + \partial v/\partial y = \partial^2\psi/\partial x\partial y - \partial^2\psi/\partial y\partial x = 0 \rightarrow \psi$ automatically satisfies continuity.
- Lines of constant ψ are streamlines (since $d\psi = \partial\psi/\partial x \, dx + \partial\psi/\partial y \, dy = -v \, dx + u \, dy = 0 \rightarrow dy/dx = v/u$).
- Flow rate between two streamlines: $q = \psi_2 - \psi_1$ (per unit depth).
- If the flow is irrotational, ψ satisfies Laplace's equation: $\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 = 0$ ($\nabla^2\psi = 0$).
- In polar coordinates: $v_r = (1/r)\,\partial\psi/\partial\theta$, $v_\theta = -\partial\psi/\partial r$.
- ψ exists for rotational and irrotational flows alike but only for 2-D incompressible flow; sign convention: flow is positive from right to left of an observer looking along increasing ψ.

Example 9

The stream function of a 2-D flow is $\psi = 3xy$. Find the velocity components, check for irrotationality, and find the discharge between the points (1, 1) and (3, 2).

Sol: $u = \partial\psi/\partial y = 3x$, $v = -\partial\psi/\partial x = -3y$. Velocity at (2, 1): $u = 6$, $v = -3$, $V = \sqrt{(36 + 9)} = 6.71$ m/s

Rotation: $\partial v/\partial x - \partial u/\partial y = 0 - 0 = 0 \rightarrow$ irrotational

$\psi_1 = 3 \times 1 \times 1 = 3$; $\psi_2 = 3 \times 3 \times 2 = 18$; $q = \psi_2 - \psi_1 = 15$ units (m^2/s)

7. Velocity Potential Function (φ)

Velocity potential

A scalar function of space and time, $\varphi(x, y, z, t)$, whose negative derivative with respect to any direction gives the velocity component in that direction. It exists only for irrotational flow.

$$u = -\partial\varphi/\partial x\,,\quad v = -\partial\varphi/\partial y\,,\quad w = -\partial\varphi/\partial z$$

Note: Sign convention: Indian textbooks (Bansal, Modi) take the negative sign. Some books use $u = +\partial\varphi/\partial x$; the physics is unchanged.

- Existence of φ requires irrotational flow ($\omega = 0$): since $\partial u/\partial y = \partial v/\partial x$ follows from equality of mixed partial derivatives of φ.
- For incompressible flow, substituting in continuity gives Laplace's equation: $\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 = 0$ ($\nabla^2\varphi = 0$). Such flows are called potential flows.
- Lines of constant φ are called equipotential lines.
- Valid for steady and unsteady flows, compressible and incompressible, 2-D and 3-D (unlike ψ, φ is defined for 3-D flow too).

Relation between ψ and φ – Flow Net

- From definitions (2-D): $\partial\varphi/\partial x = -\partial\psi/\partial y$ and $\partial\varphi/\partial y = \partial\psi/\partial x$ (Cauchy–Riemann equations).
- Slope of a streamline: $(dy/dx)_\psi = v/u$; slope of an equipotential line: $(dy/dx)_\varphi = -u/v$. Product $= -1 \rightarrow$ equipotential lines are orthogonal to streamlines.
- A grid of ψ = const and φ = const lines (equal increments) is called a flow net, used to graphically study flow around bodies, under dams, seepage, etc.

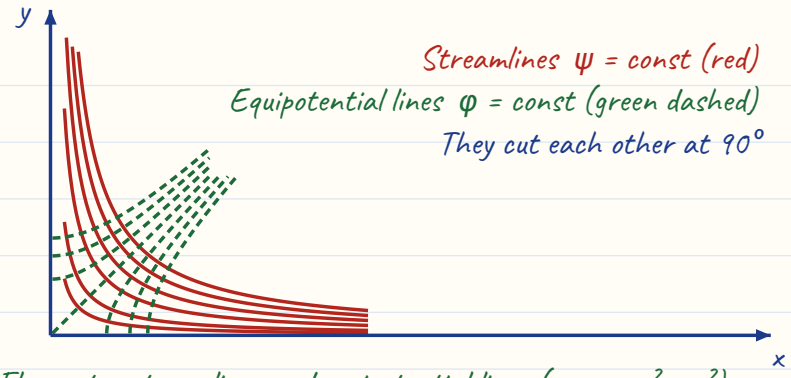


Fig 3.5: Flow net – streamlines and equipotential lines (= xy, $x^2 - y^2$)

Stream function ψ	Velocity potential φ
Defined for 2-D incompressible flow only	Defined for 2-D and 3-D flow (compressible or not)
Exists for rotational and irrotational flows	Exists only for irrotational flow
ψ = const → streamline	φ = const → equipotential line
Satisfies continuity automatically; Laplace's equation only if irrotational	Satisfies Laplace's equation if flow is incompressible (continuity)
$u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$	$u = -\partial\varphi/\partial x$, $v = -\partial\varphi/\partial y$
Difference of ψ = discharge	Gradient of φ = velocity

Example 10 – ψ and φ for a given velocity field

For the flow $u = 3x$, $v = -3y$ find (a) the stream function and (b) the velocity potential.

Sol: (a) ψ: $\partial\psi/\partial y = u = 3x \rightarrow \psi = 3xy + f(x)$; $-\partial\psi/\partial x = v = -3y \rightarrow 3y + f'(x) = 3y \rightarrow f' = 0$. $\psi = 3xy$

(b) φ: $-\partial\varphi/\partial x = 3x \rightarrow \varphi = -3x^2/2 + g(y)$; $-\partial\varphi/\partial y = -3y \rightarrow \partial\varphi/\partial y = 3y \rightarrow g = 3y^2/2$. $\varphi = -(3/2)(x^2 - y^2)$

Check orthogonality: streamline slope = $v/u = -y/x$; equipotential slope = $-u/v = x/y$; product = −1 ✓

Example 11 – Velocity from φ

A velocity potential is given by $\varphi = -(x^3/3 - x\,y^2)$. Find the velocity at (2, 3).

Sol: $u = -\partial\varphi/\partial x = x^2 - y^2$; $v = -\partial\varphi/\partial y = -2xy$. At (2, 3): $u = 4 - 9 = -5$ m/s; $v = -12$ m/s; $V = \sqrt{(25 + 144)} = 13$ m/s

Continuity: $\partial u/\partial x + \partial v/\partial y = 2x - 2x = 0$ ✓ (flow is possible and irrotational).

Example 12 – Stream function from a given velocity field

Find the stream function for the flow $u = x^2 - y^2$, $v = -2xy$, and show that it satisfies Laplace's equation.

Sol: $\partial\psi/\partial y = x^2 - y^2 \rightarrow \psi = x^2y - y^3/3 + f(x)$. Then $v = -\partial\psi/\partial x = -2xy - f'(x) = -2xy \rightarrow f' = 0$.

$\psi = x^2y - y^3/3$. $\partial^2\psi/\partial x^2 = 2y$; $\partial^2\psi/\partial y^2 = -2y$; sum = 0 ✓ (irrotational, Laplace's equation satisfied)

Summary of Fluid Kinematics

- Eulerian approach – velocity field $V(x, y, z, t)$; acceleration = local + convective.
- Steady flow: $\partial/\partial t = 0$. Uniform flow: $\partial/\partial s = 0$. Streamline = pathline = streakline in steady flow.
- Continuity (incompressible): $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$; 1-D: $A_{1V1} = A_{2V2}$.
- $\psi \rightarrow$ 2-D incompressible; $\varphi \rightarrow$ irrotational; ψ and φ lines are orthogonal and both satisfy Laplace's equation for irrotational incompressible 2-D flow.

Important Questions

1. Distinguish between Lagrangian and Eulerian approaches. Derive the expression for acceleration of a fluid particle.
2. Explain with examples: steady/unsteady, uniform/non-uniform, laminar/turbulent, rotational/irrotational flows.
3. Define streamline, pathline, streakline and stream tube. When do they coincide?
4. Derive the continuity equation in three-dimensional Cartesian coordinates. Deduce the 1-D and 2-D forms.
5. Define stream function and velocity potential. Show that streamlines and equipotential lines are orthogonal.
6. Check if a given velocity field satisfies continuity / is irrotational; find ψ and φ.

FLUID MECHANICS

Subject Code: 23A01303

UNIT - 4: Fluid Dynamics

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Surface and body forces; equations of motion – Euler's equation; Bernoulli's equation – derivation; energy principle; practical applications of Bernoulli's equation: venturimeter, orifice meter and Pitot tube; momentum principle; forces exerted by fluid flow on pipe bend; vortex flow – free and forced; definitions of Reynolds number, Froude number, Mach number, Weber number and Euler number.

1. Forces Acting on a Fluid in Motion

Newton's second law applied to a fluid particle: $F = m\,a$. The forces acting are classified as:

Type	Meaning	Examples
Surface forces	Forces acting on the boundary (surface) of a fluid element by the surroundings; proportional to area	Pressure force F_p, viscous (shear) force F_v, surface tension F_s
Body forces	Forces acting on every mass element throughout the volume; proportional to mass/volume	Gravity $F_g = mg$, magnetic and electric forces, centrifugal force
Inertia force	Rate of change of momentum of the particle (mass × acceleration)	$F_i = m\,a$

- Forces influencing fluid motion in practice: gravity, pressure, viscosity, turbulence, surface tension and compressibility.
- Net force $F_x = (F_g)_x + (F_p)_x + (F_v)_x + (F_t)_x + (F_s)_x + (F_c)_x = m\,a_x$ – Reynolds equation of motion (all forces).
- Neglecting turbulence, surface tension and compressibility: $F_g + F_p + F_v = m\,a \rightarrow$ Navier–Stokes equation.
- Neglecting also viscosity (ideal fluid): $F_g + F_p = m\,a \rightarrow$ Euler's equation of motion.

2. Euler's Equation of Motion

Consider a small cylindrical element of length ds and cross-section dA along a streamline in steady flow of an ideal fluid. Let θ be the inclination of the streamline with the horizontal.

Fluid Dynamics

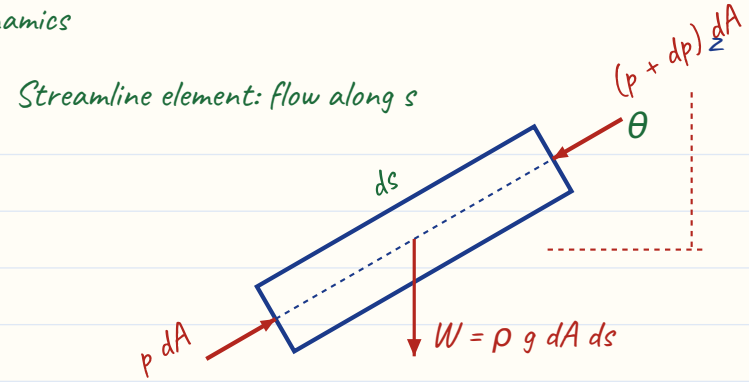


Fig 4.1: Forces on a fluid element along a streamline

- Pressure force on the upstream face = $p\,dA$; on the downstream face = $(p + dp)\,dA$ (opposing).
- Weight component along the streamline = $\rho\,g\,dA\,ds\cos\theta = \rho\,g\,dA\,dz$ (since $ds\cos\theta = dz$).
- Mass of the element = $\rho\,dA\,ds$; acceleration along s: $a_s = V\,dV/ds$ (steady flow).

Newton's second law along s: $p\,dA - (p + dp)\,dA - \rho\,g\,dA\,ds\cos\theta = \rho\,dA\,ds \cdot V\,dV/ds$

$$-dp\,dA - \rho\,g\,dA\,dz = \rho\,dA\,V\,dV$$
$$dp/\rho + g\,dz + V\,dV = 0 \text{ (Euler's equation)}$$

- Valid for steady flow along a streamline of an ideal (inviscid) fluid; it holds for compressible as well as incompressible fluid (ρ may vary with p).
- In 3-D form (Cartesian): $-(1/\rho)\,\partial p/\partial x + X = du/dt$, etc., where X, Y, Z are body force components per unit mass.

3. Bernoulli's Equation

For incompressible flow (ρ = constant), integrate Euler's equation along the streamline:

$$\int dp/\rho + g\int dz + \int V\,dV = \text{const} \rightarrow p/\rho + g\,z + V^2/2 = \text{const}$$
$$\text{Dividing by } g\text{: } p/(\rho\,g) + V^2/(2g) + z = \text{constant} = H$$

Bernoulli's Theorem
For a steady, irrotational (frictionless), incompressible flow of an ideal fluid, the total energy per unit weight (sum of pressure head, velocity head and datum head) remains constant along a streamline.

Term	Name	Meaning	Unit
$p/(\rho g)$	Pressure head	Flow work (pressure energy) per unit weight	m
$V^2/(2g)$	Velocity (kinetic) head	Kinetic energy per unit weight	m

z	Datum (potential) head	Potential energy per unit weight	m
H	Total head	Total energy per unit weight	m

- Between two sections 1 and 2: $p_1/\rho g + V_1^2/2g + z_1 = p_2/\rho g + V_2^2/2g + z_2$
- In terms of pressure: $p + \tfrac{1}{2}\rho V^2 + \rho g z = \text{const}$ (static + dynamic + hydrostatic pressure).
- In terms of energy per unit mass: $p/\rho + V^2/2 + g z = \text{const}$ (J/kg).

Assumptions in Bernoulli's Equation

1. Fluid is ideal (no viscosity, no shear stress, no friction loss).
2. Flow is steady.
3. Flow is incompressible (ρ = const).
4. Flow is along a streamline (for irrotational flow, the constant is the same throughout the field).
5. Only gravity and pressure forces act (no heat transfer, no shaft work, no turbulence or surface tension).
6. Flow is continuous with no addition or removal of energy between the sections.

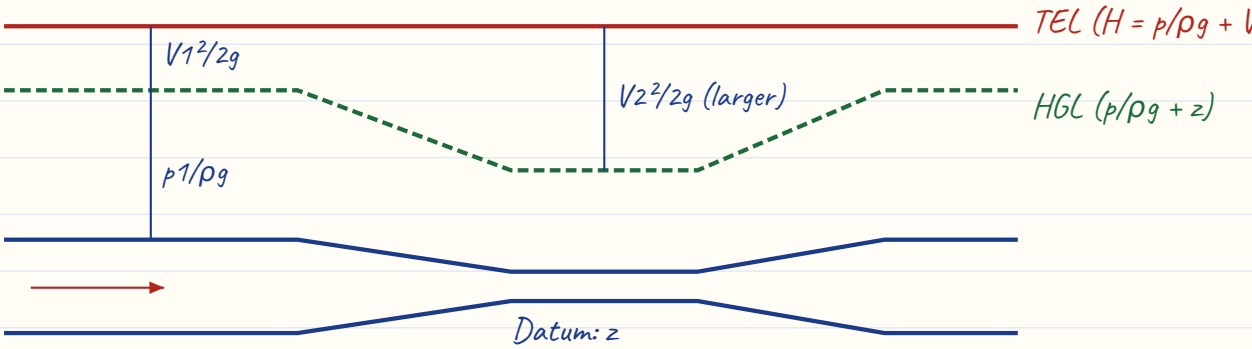


Fig 4.2: Bernoulli's equation – total energy line is horizontal for an ideal fluid

Modified Bernoulli Equation – Energy Principle for Real Fluids

For real fluids, some energy is lost due to friction (h_ℓ); a pump adds head (h_p) and a turbine extracts head (h_t). The first law of thermodynamics for steady flow gives the energy equation:

$$p_1/\rho g + \alpha_1 V_1^2/2g + z_1 + h_p = p_2/\rho g + \alpha_2 V_2^2/2g + z_2 + h_\ell + h_t$$

- α = kinetic energy correction factor (= 1 for uniform velocity, 2 for laminar pipe flow, ≈ 1.05–1.1 for turbulent flow; usually taken as 1).
- Power of a pump or turbine: $P = \rho g Q H$ (watt); shaft power = P/η.
- Bernoulli's equation is a statement of conservation of mechanical energy; the energy equation extends it to include losses and machines.

Example 1 – Bernoulli's equation

Water flows at 0.04 m^3/s through a pipe that reduces from 200 mm (section 1, gauge pressure 50 kPa, elevation 0) to 100 mm (section 2, elevation 3 m above section 1). Neglecting losses find p_2.

Sol: $V_1 = Q/A_1 = 0.04/0.03142 = 1.273$ m/s; $V_2 = 0.04/0.007854 = 5.093$ m/s

$p_1/\rho g = 50000/9810 = 5.097$ m; $V_1^2/2g = 0.0826$ m; $H = 5.097 + 0.083 + 0 = 5.179$ m

$p_2/\rho g = H - V_2^2/2g - z_2 = 5.179 - 1.322 - 3 = 0.857$ m $\rightarrow p_2 = 0.857 \times 9810 = 8.41$ kPa (gauge)

Example 2 – Pump power (energy equation)

A pump lifts 0.05 m^3/s of water from a sump to a tank 20 m above; total head loss in pipes is 3 m. Find the pump head and the shaft power if $\eta = 70\%$.

Sol: Head to be supplied $H = 20 + 3 = 23$ m. Water power $= \rho g Q H = 9810 \times 0.05 \times 23 = 11281\ W \approx 11.28$ kW

Shaft power $= 11.28/0.70 = 16.1$ kW

4. Practical Applications of Bernoulli's Equation

(a) Venturimeter

A device used to measure the discharge in a pipe. It consists of (i) a converging cone (angle ≈ 20°), (ii) a short cylindrical throat, and (iii) a gradually diverging cone (angle 5°–7°) which recovers pressure with low loss.

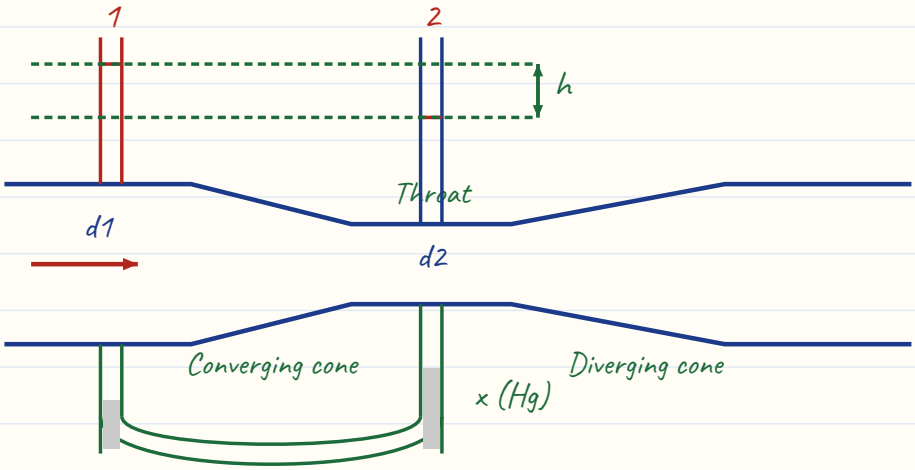


Fig 4.3: Venturimeter with differential manometer

Apply Bernoulli between section 1 (inlet, area a_1) and section 2 (throat, area a_2) for a horizontal meter ($z_1 = z_2$), ideal flow:

$$p_1/\rho g + V_1^2/2g = p_2/\rho g + V_2^2/2g \rightarrow h = (p_1 - p_2)/\rho g = (V_2^2 - V_1^2)/2g$$

From continuity $a_1 V_1 = a_2 V_2 \rightarrow V_1 = a_2 V_2/a_1$

$$Q_{th} = a_1 a_2 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)}$$

$$Q_{act} = C_d \cdot a_1 a_2 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)}$$

- h = differential pressure head (m of the flowing liquid) = $x (S_m/S_f - 1)$ for a U-tube differential manometer with manometric liquid of S_m, x = reading. For mercury and water: $h = 12.6\,x$.
- For a manometer with a lighter liquid (inverted U-tube): $h = x (1 - S/S_f)$.
- C_d = coefficient of discharge (0.96 – 0.98) accounts for losses; always < 1.
- Inclined / vertical venturimeter: the same formula holds, but $h = (p_1/\rho g + z_1) - (p_2/\rho g + z_2)$ – the piezometric head difference, read directly from the differential manometer.
- Merits: accurate, low head loss, self-cleaning; Demerits: costly, long, not flexible.

Example 3 – Venturimeter

A horizontal venturimeter has inlet 300 mm and throat 150 mm diameter. A mercury differential manometer reads 0.2 m. Find the discharge for $C_d = 0.98$.

Sol: $h = x (S_m/S - 1) = 0.2 \times 12.6 = 2.52$ m of water. $a_1 = 0.07069\ m^2$; $a_2 = 0.01767\ m^2$

$\sqrt{(a_1^2 - a_2^2)} = \sqrt{(0.0049966 - 0.0003123)} = 0.06844$; $\sqrt{(2gh)} = \sqrt{(2 \times 9.81 \times 2.52)} = 7.031$

$Q = 0.98 \times (0.07069 \times 0.01767) \times 7.031/0.06844 = 0.1258\ m^3/s \approx 125.8\ L/s$

(b) Orifice Meter

A thin plate with a sharp-edged circular hole (diameter $d_0 \approx 0.5\, d_1$) fixed concentrically in the pipe between flanges. As the fluid passes, the jet contracts to a minimum area called the vena contracta (section 2, $a_c = C_c\, a_0$), where the velocity is maximum and pressure minimum. Pressure difference between a section upstream (1) and the vena contracta (2) is measured by a differential manometer.

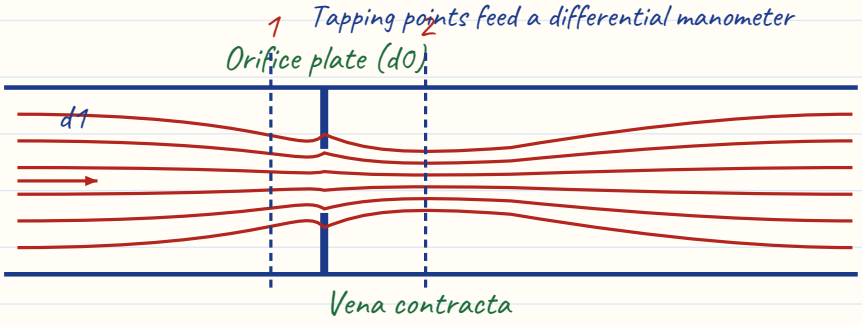


Fig 4.4: Orifice meter

$$Q = C_d \cdot a_0 a_1 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_0^2)}$$

- Derived like the venturimeter (Bernoulli + continuity) with C_d including the contraction coefficient ($C_d \approx 0.6 - 0.65$, much lower than for a venturimeter).
- Merits: cheap, compact, easy to install. Demerits: large permanent pressure (head) loss (since jet expansion causes eddies), lower accuracy, clogging by suspended particles.

Example 4 – Orifice meter

A 200 mm pipe has a 100 mm sharp-edged orifice ($C_d = 0.6$). A mercury manometer reads 0.15 m. Find the discharge of water.

Sol: $h = 0.15 \times 12.6 = 1.89$ m; $a_1 = 0.031416$, $a_0 = 0.007854$ m^2; $\sqrt{(a_1^2 - a_0^2)} = 0.030418$; $\sqrt{(2gh)} = 6.0895$

$Q = 0.6 \times (0.031416 \times 0.007854) \times 6.0895/0.030418 = 0.0296$ m^3/s ≈ 29.6 L/s

Feature	Venturimeter	Orifice meter
Coefficient of discharge	0.96 – 0.98	0.60 – 0.65
Head loss	Small (gradual diverging cone)	Large (sudden expansion)
Cost / size	Costly, long	Cheap, compact
Accuracy	High	Moderate

(c) Pitot Tube

A device to measure the local velocity at a point in a flow. A glass tube bent at 90° is placed with its open end facing the flow. The fluid at the open end is brought to rest (stagnation point) and rises in the vertical limb by height h above the free surface (or static pressure level).

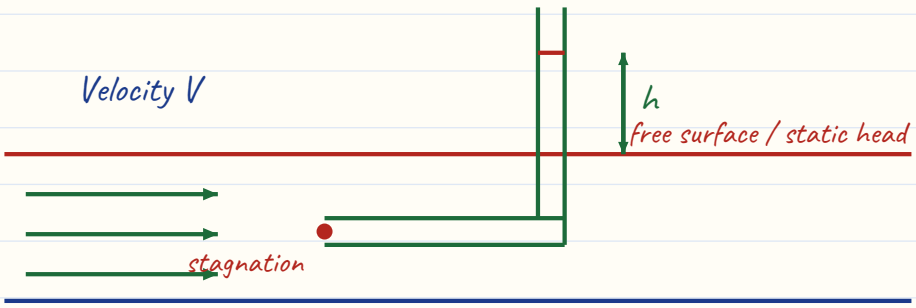


Fig 4.5: Pitot tube in an open channel

Apply Bernoulli between a point 1 upstream (velocity V, static head $p/\rho g$) and the stagnation point 2 ($V_2 = 0$):

$$p/\rho g + V^2/2g = p_0/\rho g \rightarrow h = V^2/2g$$
$$V = \sqrt{(2 g h)} \text{ and actual } V = C_v \sqrt{(2 g h)}$$

- p_0 = stagnation pressure = static pressure + dynamic pressure ($\frac{1}{2}\rho V^2$). C_v = coefficient of the pitot tube ≈ 0.98–1.0.
- Pitot-static tube: a concentric tube arrangement, the inner tube senses stagnation pressure and the outer one with side holes senses static pressure. Differential manometer reads h: $V = C_v \sqrt{(2 g h)}$ with h in metres of flowing fluid.
- For gas flow with a liquid manometer: h = (ρ_manometric/ρ_gas) × x. Used in aircraft airspeed indicators and ducts.

Example 5 – Pitot-static tube in air duct

A pitot-static tube in an air duct (ρ = 1.2 kg/m^3) shows a water-manometer deflection of 25 mm. Find the air velocity (C_v = 1).

Sol: h_{air} = (1000/1.2) × 0.025 = 20.83 m of air; $V = \sqrt{(2 \times 9.81 \times 20.83)}$ = 20.2 m/s

5. Momentum Principle

Impulse–Momentum Equation

The net external force acting on a control volume of fluid in a given direction equals the rate of change of momentum of the fluid in that direction (outflow of momentum minus inflow of momentum).

$$\Sigma F = d(mV)/dt = \rho Q (V_2 - V_1) \text{ (steady flow, one inlet/outlet)}$$
$$\Sigma F_x = \rho Q (V_{2x} - V_{1x}), \Sigma F_y = \rho Q (V_{2y} - V_{1y})$$

- ΣF includes pressure forces on the control surfaces, body forces (weight) and the force exerted by the solid boundary on the fluid.
- The force exerted by the fluid on the solid is equal and opposite to that exerted by the solid on the fluid.
- Momentum correction factor $\beta = (1/A V^2) \int v^2 dA$: = 1 for uniform flow; 1.33 for laminar pipe flow; 1.02–1.05 for turbulent flow.
- Used when the force is required but the energy losses in the flow are unknown (pipe bends, nozzles, jets striking plates, hydraulic jump, turbines).

Force Exerted by Flowing Fluid on a Pipe Bend

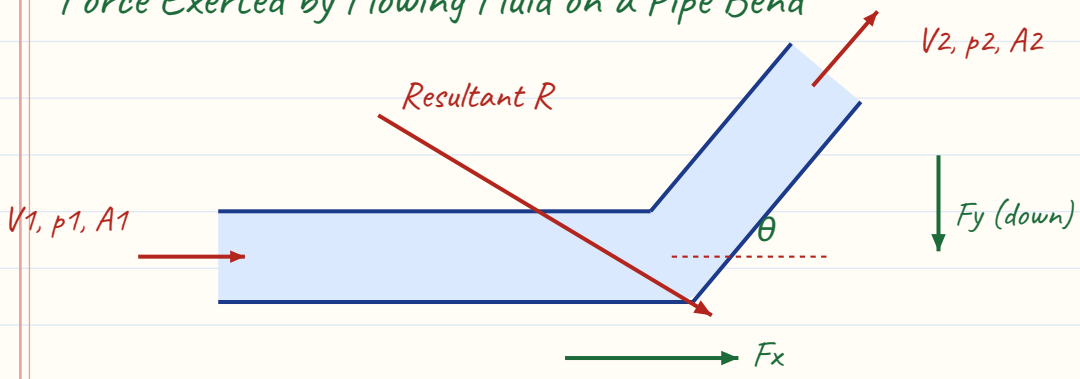


Fig 4.6: Forces on a horizontal pipe bend (deflection)

A pipe bend turns the flow through angle θ, the section changes from A_1 to A_2. Let p_1, p_2 be the gauge pressures. Take x along the inlet flow direction. By continuity $V_1 = Q/A_1$ and $V_2 = Q/A_2$; p_2 follows from Bernoulli's equation (with losses).

Forces on the fluid in the control volume (x-direction): $p_1 A_1 - p_2 A_2 \cos\theta + R_x = \rho Q (V_2 \cos\theta - V_1)$. Force by fluid on the bend is $F_x = -R_x$:

$$F_x = \rho Q (V_1 - V_2 \cos\theta) + p_1 A_1 - p_2 A_2 \cos\theta$$

$$F_y = \rho Q V_2 \sin\theta + p_2 A_2 \sin\theta \text{ (acts opposite to the y-component of outflow)}$$

$$F_R = \sqrt{(F_x^2 + F_y^2)},\ \tan\alpha = F_y / F_x$$

- α = inclination of the resultant force with the x-axis (direction of inlet flow).
- If the bend is in a vertical plane, add the weight of the fluid and bend in the y-direction.
- Large forces on bends of water mains/penstocks need anchor blocks / thrust blocks to resist F_R.
- For a 90° bend of uniform diameter: $V_1 = V_2 = V$, $p_1 \approx p_2 = p$: $F_x = p A$ (and $F_y = \rho Q V + p A$).

Example 6 – Reducing pipe bend

Water flows at 0.1 m^3/s through a horizontal 60° bend reducing from 300 mm to 150 mm. Inlet gauge pressure is 100 kPa. Neglecting losses, find the force on the bend.

Sol: $A_1 = 0.07069\ m^2$; $A_2 = 0.017671\ m^2$; $V_1 = 1.415\ m/s$; $V_2 = 5.659\ m/s$; $\rho Q = 100\ kg/s$

Bernoulli: $p_2 = p_1 + \frac{1}{2}\rho(V_1^2 - V_2^2) = 100000 + 500\ (2.001 - 32.02) = 84990\ Pa = 85.0\ kPa$

$p_{1A1} = 7068.6\ N$; $p_{2A2} = 84990 \times 0.017671 = 1501.9\ N$

$F_x = 100\ (1.415 - 5.659 \times 0.5) + 7068.6 - 1501.9 \times 0.5 = -141.5 + 7068.6 - 751.0 = 6176\ N$

$F_y = (\rho Q V_2 + p_{2A2}) \sin 60° = (565.9 + 1501.9) \times 0.866 = 1791\ N$

$F_R = \sqrt{(6176^2 + 1791^2)} = 6431\ N$; $\alpha = \tan^{-1}(1791/6176) = 16.2°$ to the inlet flow direction.

6. Vortex Flow

A vortex flow is a flow of fluid along curved (concentric circular) paths about an axis. Two types:

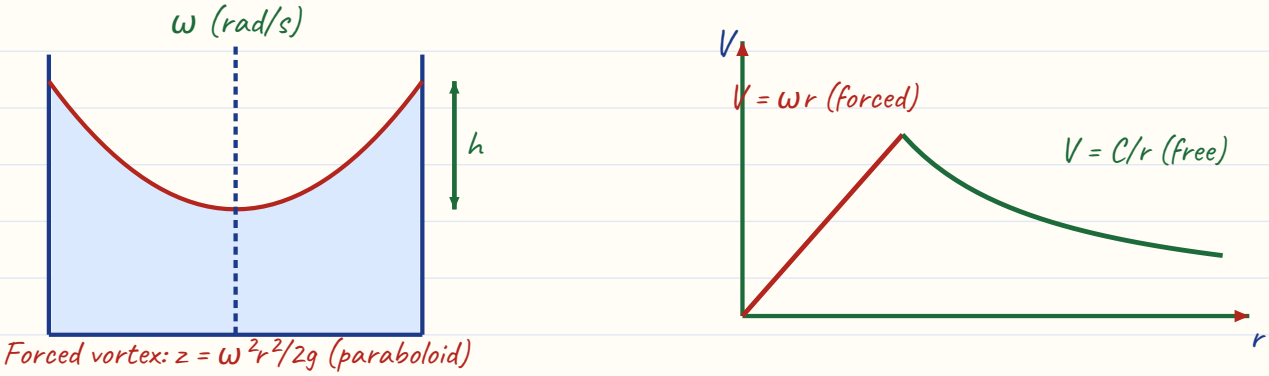


Fig 4.7: Forced vortex (left) and velocity distribution of free and forced vortices (right)

(a) Forced Vortex (rotational, 'solid-body' rotation)

Fluid is rotated by an external torque such that every particle has the same angular velocity ω, e.g. liquid in a cylinder rotated about its vertical axis, centrifugal pump impeller.

- Tangential velocity $V = \omega r$ (increases linearly with r); flow is rotational (vorticity = 2ω).
- Pressure variation: $\partial p/\partial r = \rho \omega^2 r$, $\partial p/\partial z = -\rho g$. Integrating: $p = p_0 + \rho \omega^2 r^2/2 - \rho g z$.
- Free-surface profile ($p = p_{atm}$): $z = \omega^2 r^2 / 2g$ – a paraboloid of revolution measured from the vertex (lowest point on the axis).
- Pressure head at the wall of a closed cylinder of radius R: $h = \omega^2 R^2/2g$ (centrifugal head).
- Volume of paraboloid of revolution = ½ × volume of circumscribing cylinder. Rise of the surface at the wall = fall at the centre = $\omega^2 R^2/4g$ (for a cylinder filled initially to half-way of the paraboloid height, with no uncovering of the bottom).

(b) Free Vortex (irrotational vortex)

Fluid rotates by itself without any external torque, e.g. whirlpool, drain vortex, cyclone, flow around a bend in a river, tornado outside the core.

- Angular momentum is conserved: $V \times r = C$ = constant → $V = C/r$ (velocity is highest near the centre, $V \to \infty$ as $r \to 0$; so a free vortex has a forced-vortex core in practice: Rankine combined vortex).
- Flow is irrotational ($\omega_z = 0$ everywhere except $r = 0$); Bernoulli's equation holds across streamlines, with constant H.
- Pressure: from Bernoulli, $p/\rho g + V^2/2g$ = const → p increases with r ($p_2 - p_1 = \frac{1}{2}\rho (V_1^2 - V_2^2)$) and the free surface profile is $z = z_\infty - C^2/(2 g r^2)$.
- Circulation $\Gamma = 2\pi r V = 2\pi C$ (constant).

Aspect	Forced vortex	Free vortex
Rotation	Rotational ($\zeta = 2\omega$)	Irrotational ($\zeta = 0$)
Velocity	$V = \omega r$ ($\propto r$)	$V = C/r$ ($\propto 1/r$)
Energy	External energy supplied	No external energy, H const
Free surface	$z = \omega^2 r^2/2g$ (paraboloid)	$z = z_\infty - C^2/2gr^2$ (hyperboloid-like)
Examples	Rotating tank, centrifuge, impeller	Whirlpool, drain, cyclone, tornado

Example 7 – Rotating cylinder

An open cylinder of radius 0.15 m and height 0.5 m contains water to a depth of 0.4 m. At what speed can it be rotated about its axis without spilling?

Sol: Spilling begins when the rise at the wall equals 0.5 – 0.4 = 0.1 m. For a paraboloid, rise at wall = fall at centre = $h/2$ where $h = \omega^2 R^2/2g$.

$h = 0.2$ m → $\omega^2 = 2 g h/R^2 = 2 \times 9.81 \times 0.2/0.0225 = 174.4$ → $\omega = 13.2$ rad/s

$N = 60\,\omega/2\pi = 126$ rpm (centre depth = 0.4 – 0.1 = 0.3 m > 0, so the bottom stays covered ✓)

7. Dimensionless Numbers (Dynamic Similarity)

Dimensionless numbers are ratios of two forces acting on a fluid element. For dynamic similarity between a model and prototype, the relevant number must be equal. Let V = velocity, L = characteristic length, ρ = density.

Number	Definition (ratio of forces)	Formula	Significance / Use

Reynolds number Re	Inertia force / Viscous force	$Re = \rho V L / \mu = V L / \nu$	Pipe flow, boundary layers, submarines, flows with viscous forces dominant
Froude number Fr	Inertia force / Gravity force	$Fr = V / \sqrt{(g L)}$	Free-surface flows: open channels, spillways, ships, hydraulic jump; Fr < 1 subcritical, = 1 critical, > 1 supercritical
Mach number M	Inertia force / Elastic (compressibility) force	$M = V / c = V/\sqrt{(K/\rho)}$	Compressible flows: aircraft, rockets, nozzles; M < 1 subsonic, = 1 sonic, > 1 supersonic
Weber number We	Inertia force / Surface-tension force	$We = \rho V^2 L / \sigma$	Capillary waves, droplets, thin films, bubbles, flow over weirs at low heads
Euler number Eu	Pressure force / Inertia force	$Eu = \Delta p/(\rho V^2) = V/\sqrt{(\Delta p/\rho)}$	Pressure-driven flows: cavitation (Eu = cavitation number), turbo-machines, pipe losses

- Reynolds model law: $(\rho V L/\mu)_m = (\rho V L/\mu)_p$ – used for enclosed flows (pipes, valves, flow around submerged bodies).
- Froude model law: $(V/\sqrt{(gL)})_m = (V/\sqrt{(gL)})_p$ – used for free-surface flows (weirs, spillways, ships).
- Speed of sound in a gas: $c = \sqrt{(k R T)}$; in a liquid: $c = \sqrt{(K/\rho)}$.

Example 8 – Dimensionless numbers

Water ($\nu = 1 \times 10^{-6}$ m²/s) flows at 2 m/s in a 50 mm pipe. Find Re. Also find the Froude number for a river flow of V = 3 m/s at depth 1.5 m, and the Mach number of an aircraft at 250 m/s in air at 15 °C.

Sol: $Re = V D/\nu = 2 \times 0.05/10^{-6} = 1 \times 10^5$ (turbulent). $Fr = 3/\sqrt{(9.81 \times 1.5)} = 3/3.836 = 0.78$ (subcritical).

$c = \sqrt{(1.4 \times 287 \times 288)} = 340$ m/s; $M = 250/340 = 0.735$ (subsonic).

Summary of Fluid Dynamics

Concept	Equation
Euler's equation	$dp/\rho + g\,dz + V\,dV = 0$
Bernoulli's equation	$p/\rho g + V^2/2g + z = const$
Venturimeter	$Q = C_d a_1 a_2 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_2^2)}$
Orifice meter	$Q = C_d a_0 a_1 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_0^2)}$
Pitot tube	$V = C_v \sqrt{(2\,g\,h)}$
Momentum equation	$F = \rho\,Q\,(V_2 - V_1)$
Force on bend	$F_x = \rho Q(V_1 - V_{2\cos}\theta) + p_{1A1} - p_{2A2\cos}\theta; F_y = (\rho Q V_2 + p_{2A2})\sin\theta$
Forced vortex	$V = \omega r, z = \omega^2 r^2/2g$
Free vortex	$V\,r = const$

Important Questions

1. Derive Euler's equation of motion and hence Bernoulli's equation. State its assumptions.
2. With a neat sketch, derive the discharge through a venturimeter. Why is its C_d higher than that of an orifice meter?
3. Explain the working of a Pitot tube and derive the velocity expression.
4. State the momentum equation and derive the force exerted by a flowing fluid on a pipe bend.
5. Differentiate between free and forced vortex. Derive the equation of the free surface in a forced vortex.
6. Define Reynolds, Froude, Mach, Weber and Euler numbers with their significance.

FLUID MECHANICS

Subject Code: 23A01303

UNIT - 5: Analysis of Pipe Flow

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Energy losses in pipelines; Darcy–Weisbach equation; minor losses in pipelines; hydraulic grade line and total energy line; concept of equivalent length – pipes in parallel and series.

1. Energy Losses in Pipelines

When a real fluid flows in a pipe, a part of its energy is dissipated as heat due to viscosity and turbulence. This is called head loss (energy loss). Total head loss in a pipeline has two parts:

Major losses (h_f)	Minor losses (h_m)
Due to friction between fluid layers and pipe wall along the full length of the pipe	Due to local disturbances: change in section, bends, valves, fittings, entry, exit
Proportional to pipe length	Occur at a point/short length
Computed by Darcy–Weisbach / Chezy / Manning formulae	Computed as $K V^2/2g$ (K = loss coefficient)
Dominant in long pipes	Can be neglected when $L/D > 1000$ (long pipes), but not in short pipes

$$h_L = h_f + h_m$$

Reynolds Number and Flow Regimes in Pipes

- $Re = \rho V D / \mu = V D / \nu$. Laminar if $Re < 2000$; transitional 2000 – 4000; turbulent if $Re > 4000$.
- Laminar flow – head loss $\propto V$; turbulent flow – head loss $\propto V^{1.75-2}$.

2. Darcy–Weisbach Equation (Major Loss)

Consider steady uniform flow through a horizontal pipe of diameter D and length L. Let p_1, p_2 be the pressures at the ends and τ_0 the average wall shear stress.

Analysis of Pipe Flow

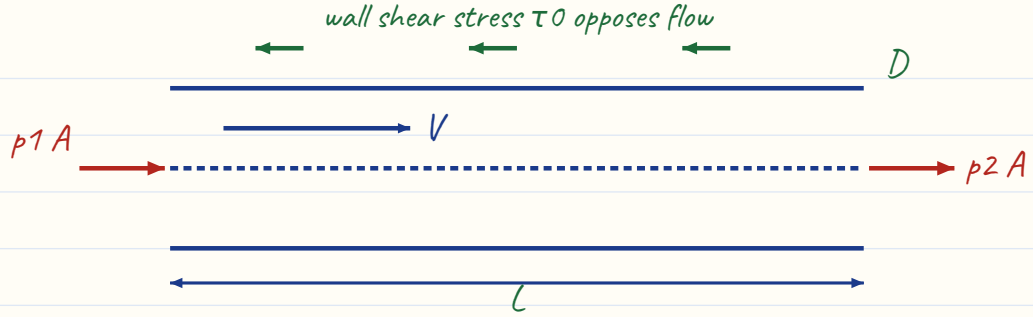


Fig 5.1: Force balance on a pipe element

Force balance in the flow direction (uniform flow – no acceleration):

$$(p_1 - p_2)(\pi D^2/4) = \tau_0 (\pi D L) \rightarrow (p_1 - p_2)/\rho g = h_f = 4\tau_0 L/(\rho g D)$$

From dimensional analysis/experiment, wall shear stress is proportional to the dynamic pressure: $\tau_0 = (f/8)\rho V^2$ where f = Darcy friction factor. Substituting:

$$h_f = f L V^2/(2 g D) \text{ (Darcy–Weisbach equation)}$$

In terms of discharge ($V = 4Q/\pi D^2$): $h_f = 8 f L Q^2/(\pi^2 g D^5)$

- h_f = loss of head due to friction (m of fluid); L = length of pipe; D = diameter; V = mean velocity; f = friction factor (dimensionless).
- Pressure drop: $\Delta p = \rho g h_f = f (L/D)(\rho V^2/2)$.
- Some books use the Fanning friction factor $f' = f/4$, giving $h_f = 4 f' L V^2/(2 g D)$ – check the definition used.
- Observation: $h_f \propto L$, $\propto V^2$ (fully rough turbulent), $\propto 1/D$ (for fixed V), $\propto Q^2/D^5$ (for fixed Q) – hence doubling the diameter reduces head loss by 32 times for the same discharge!
- Power lost in friction = $\rho g Q h_f$

Friction Factor f

Flow condition	Expression for f
Laminar (Re < 2000)	$f = 64/Re$ (Hagen–Poiseuille); independent of roughness
Turbulent, smooth pipes, 4000 < Re < 10^5	$f = 0.316 / Re^{0.25}$ (Blasius)
Turbulent, smooth pipes, any Re	$1/\sqrt{f} = 2 \log_{10}(Re\sqrt{f}) - 0.8$ (Prandtl–von Kármán)
Fully rough turbulent flow	$1/\sqrt{f} = 2 \log_{10}(D/\varepsilon) + 1.14$ (von Kármán–Nikuradse)
Transition (commercial pipes)	$1/\sqrt{f} = -2 \log_{10}[\varepsilon/(3.7 D) + 2.51/(Re\sqrt{f})]$ (Colebrook–White)

Analysis of Pipe Flow

- Hagen–Poiseuille (laminar): $h_f = 32\,\mu V L / (\rho g D^2) \rightarrow \Delta p = 32\,\mu V L/D^2$ (obtained by putting $f = 64/Re$ in Darcy's equation).
- Relative roughness ε/D: ε (absolute roughness) – commercial steel 0.045 mm; wrought iron 0.046 mm; galvanised iron 0.15 mm; cast iron 0.26 mm; concrete 0.3 – 3 mm; PVC/drawn tubing ≈ 0.0015 mm.
- In turbulent flow f depends on Re and ε/D; for large Re, f depends on ε/D only (the curves flatten).

Moody Diagram

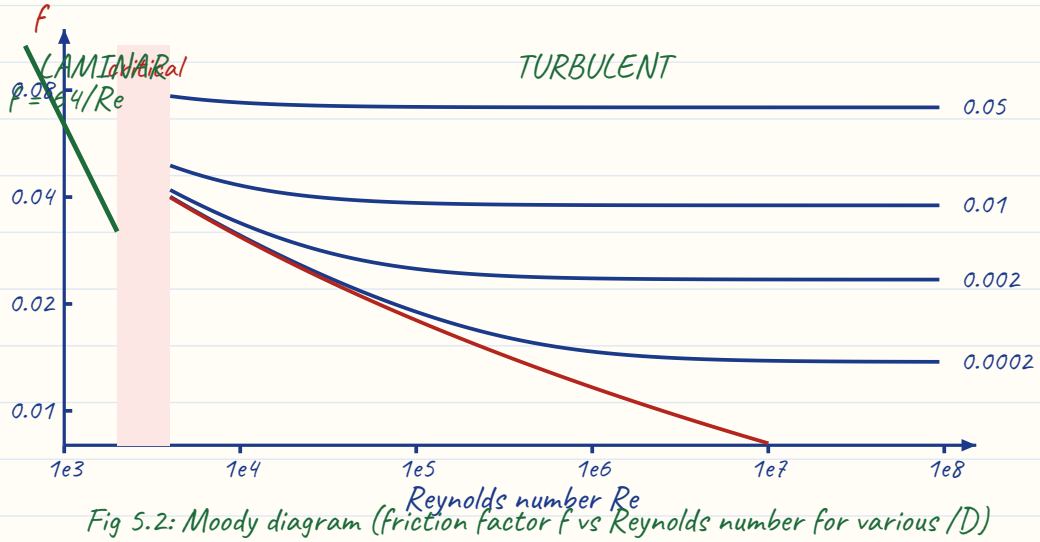


Fig 5.2: Moody diagram (friction factor f vs Reynolds number for various /D)

- Plot of f against Re (log–log) for various ε/D; it gives f for any pipe. Laminar region: straight line $f = 64/Re$. Critical zone ($2000 < Re < 4000$): uncertain. Transition: curves fall with Re. Fully rough: horizontal lines.
- Procedure: calculate Re and ε/D; read f from the chart; then $h_f = f\,L\,V^2/2g\,D$.

Other Empirical Formulae (for open and closed channels)

- Chezy's formula: $V = C\sqrt{(m\,i)}$, where m = hydraulic mean depth (hydraulic radius) = $A/P = D/4$ for a circular pipe running full, $i = h_f/L$ (slope), C = Chezy's constant. Then $C = \sqrt{(8g/f)}$.
- Manning's formula: $V = (1/N)\, m^{2/3}\, i^{1/2}$ (N = Manning roughness coefficient).
- Hazen–Williams: $V = 0.849\, C_{HW}\, R^{0.63}\, S^{0.54}$ (water supply pipes; $C_{HW} \approx 100$–140).

Example 1 – Head loss by Darcy–Weisbach

Water flows at 0.05 m^3/s through a 500 m long, 200 mm diameter pipe, $f = 0.02$. Find the head loss and the pressure drop.

Sol: $V = Q/A = 0.05/0.031416 = 1.5915$ m/s; $V^2/2g = 2.533/19.62 = 0.1291$ m

$h_f = f\,(L/D)\,V^2/2g = 0.02 \times (500/0.2) \times 0.1291 = 6.455$ m

$\Delta p = \rho g h_f = 9810 \times 6.455 = 63.3$ kPa; Power lost $= 9810 \times 0.05 \times 6.455 = 3.17$ kW

Example 2 – Laminar flow of oil

Oil ($S = 0.9$, $\mu = 0.1$ N·s/m²) flows at 1 m/s through a 50 mm pipe, 100 m long. Check the type of flow and find the head loss and pressure drop.

Sol: $Re = \rho VD/\mu = 900 \times 1 \times 0.05/0.1 = 450 < 2000 \rightarrow$ laminar. $f = 64/Re = 0.1422$

$h_f = f\,L\,V^2/(2gD) = 0.1422 \times 100 \times 1/(19.62 \times 0.05) = 14.5$ m of oil (check: $32\,\mu VL/(\rho g D^2) = 14.5$ m ✓)

$\Delta p = \rho\, g\, h_f = 900 \times 9.81 \times 14.5 = 128$ kPa

Example 2(b) – Blasius formula

Water ($\nu = 10^{-6}$ m²/s) flows at 1 m/s in a smooth 100 mm pipe of length 200 m. Find f and the head loss.

Sol: $Re = V\,D/\nu = 1 \times 0.1/10^{-6} = 1 \times 10\text{^}5$ (turbulent). Blasius: $f = 0.316/Re^{0.25} = 0.316/17.78 = 0.01777$

$h_f = f\,(L/D)\,V^2/2g = 0.01777 \times 2000 \times 0.05097 = 1.81$ m

3. Minor Losses in Pipelines

Minor losses arise from sudden changes in flow geometry which cause separation, eddies and mixing. They are expressed as:

$$h_m = K\,V^2 / 2g \quad (K = \text{loss coefficient, found by experiment})$$

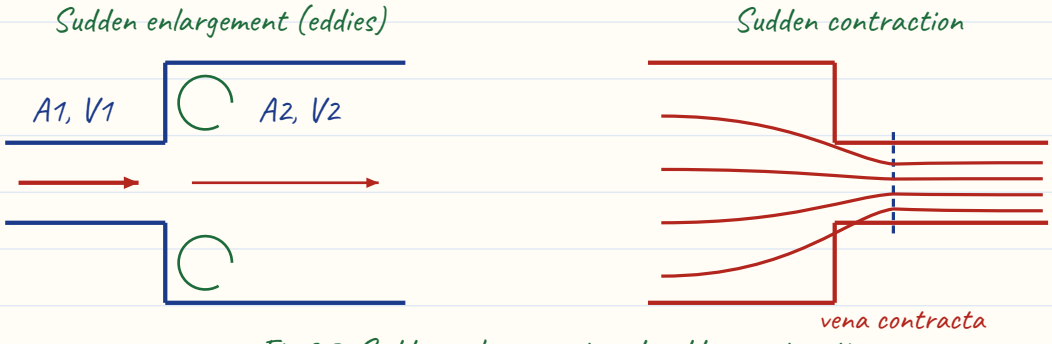


Fig 5.3: Sudden enlargement and sudden contraction

(a) Loss due to Sudden Enlargement

Applying the momentum equation between section 1 (just after the enlargement, where pressure acts on the full area A_2) and section 2, together with the energy and continuity equations (Borda–Carnot):

$$h_e = (V_1 - V_2)^2 / 2g = (V_1^2/2g)(1 - A_1/A_2)^2$$

- Pressure rise across the enlargement: $(p_2 - p_1)/\rho g = (V_1^2 - V_2^2)/2g - h_e = V_2 (V_1 - V_2)/g$.
- Exit loss from a pipe into a large reservoir ($A_2 \to \infty$, $V_2 = 0$): $h_e = V^2/2g$ ($K = 1$).

(b) Loss due to Sudden Contraction

The flow contracts to a vena contracta (area $A_c = C_c A_2$) and then expands to fill the smaller pipe; the loss is mainly the sudden-enlargement loss from the vena contracta to section 2:

$$h_c = (1/C_c - 1)^2 V_2^2/2g = K_c V_2^2/2g \ (C_c \approx 0.62 \to K_c \approx 0.375; \text{commonly } h_c = 0.5 V_2^2/2g)$$

(c) Other Minor Losses

Source of loss	Loss head	Typical K
Entrance (sharp-edged, from reservoir)	$0.5 V^2/2g$	0.5 (rounded: 0.05 – 0.2; re-entrant: 0.8)
Exit (pipe to reservoir)	$V^2/2g$	1.0
Gradual enlargement (cone)	$k (V_1 - V_2)^2/2g$	k = 0.1 – 0.5 depending on angle
Gradual contraction	$k V_2^2/2g$	0.02 – 0.05
Pipe bend / elbow 90°	$K V^2/2g$	0.9 (standard), 0.3 (long radius)
Elbow 45°	$K V^2/2g$	0.4
Gate valve (fully open)	$K V^2/2g$	0.15 – 0.2 (half open ≈ 5.6)
Globe valve (fully open)	$K V^2/2g$	≈ 10
Check (swing) valve	$K V^2/2g$	≈ 2.5
Loss due to an obstruction (partial blockage)	$(A/(C_c(A - a)) - 1)^2 V^2/2g$	–

Note: Total head loss of a pipeline of length L and diameter D with minor items: $h_L = (f L/D + \Sigma K) V^2/2g$.

Example 3 – Sudden enlargement

A pipe enlarges suddenly from 150 mm to 300 mm diameter. Discharge is 0.08 m^3/s. Find the head loss and the pressure rise.

Sol: $V_1 = 0.08/0.017671 = 4.527$ m/s; $V_2 = 0.08/0.070686 = 1.132$ m/s

$h_e = (V_1 - V_2)^2/2g = (3.395)^2/19.62 = 0.588$ m

$(p_2 - p_1)/\rho g = (V_1^2 - V_2^2)/2g - h_e = (20.49 - 1.28)/19.62 - 0.588 = 0.979 - 0.588 = 0.392$ m $\rightarrow \Delta p = 3.84$ kPa

Example 4 – Sudden contraction

A 300 mm pipe suddenly contracts to 150 mm; $Q = 0.05\ m^3/s$. Find the loss (use $h_c = 0.5\ V_2^2/2g$).

Sol: $V_2 = 0.05/0.017671 = 2.829$ m/s; $h_c = 0.5 \times (2.829^2/19.62) = 0.5 \times 0.408 = 0.204$ m

4. Equivalent Length (Concept)

A fitting (valve, bend, etc.) can be replaced by an equivalent length L_e of straight pipe of the same diameter that gives the same head loss. Equating minor loss and friction loss:

$$K\,V^2/2g = f\,L_e\,V^2/(2\,g\,D) \rightarrow L_e = K\,D/f$$

- The total 'virtual' pipe length $L_{total} = L + \Sigma L_e$ can be used in Darcy's equation with a single f.
- Equivalent length is often quoted in terms of diameters: $L_e/D = K/f$ (e.g., a globe valve ≈ 340 D).
- Not to be confused with the equivalent pipe (next section) – which replaces a whole compound pipe system by a single pipe.

Example 5

A globe valve ($K = 10$) is in a 100 mm pipe with $f = 0.025$. Find its equivalent length.

Sol: $L_e = K\,D/f = 10 \times 0.1/0.025 = 40$ m $(= 400\ D)$

5. Hydraulic Grade Line (HGL) and Total Energy Line (TEL)

Total Energy Line (TEL) / Energy Grade Line

A line which joins the total head at every section of a pipe: $p/\rho g + V^2/2g + z$ measured above a datum.

Hydraulic Grade Line (HGL) / Piezometric Line

A line which joins the piezometric head at every section: $p/\rho g + z$. It shows the level to which water would rise in a piezometer tube fixed to the pipe.

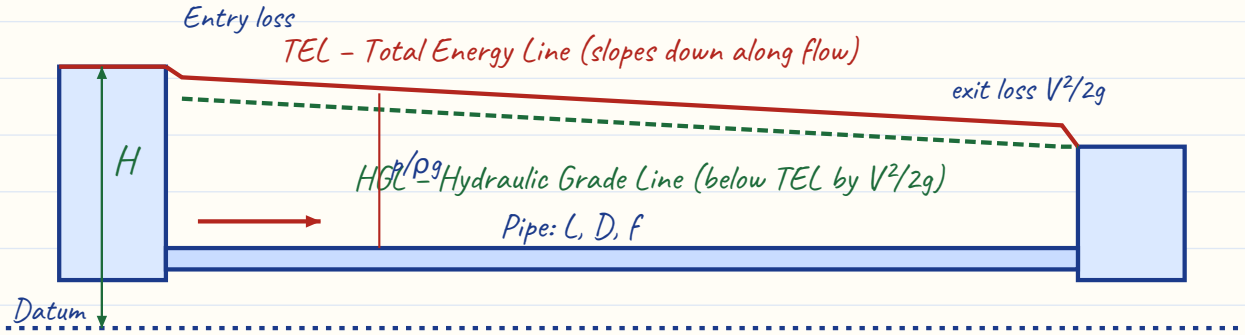


Fig 5.4: TEL and HGL for a pipe connecting two reservoirs

Properties of TEL and HGL

- TEL is always above HGL by the velocity head $V^2/2g$. For a pipe of uniform diameter the two are parallel.
- TEL always slopes downward in the direction of flow because of head losses (it rises only where a pump adds head; it falls sharply at a turbine). The slope = h_f/L = hydraulic gradient.
- HGL may rise or fall: it rises in a diverging section (velocity falls, pressure recovers) and falls in a converging section.
- At the free surface of a reservoir, TEL and HGL coincide with the water surface ($V = 0$, $p = 0$ gauge). At a free jet discharge to the atmosphere HGL passes through the centre of the outlet.
- Sudden drop in both lines occurs at local losses (entrance, valve, sudden enlargement).
- If the pipe rises above the HGL, the pressure in that portion is below atmospheric (negative gauge), e.g. at the crest of a siphon – risk of cavitation / air release. If the pipe lies below the HGL, the pressure is positive.
- Vertical distance from pipe centre-line to the HGL = pressure head at that point.
- For a long pipe where minor losses and velocity head are neglected, HGL ≈ TEL.

Example 6 – Flow between two reservoirs with minor losses

Two reservoirs with a level difference of 12 m are connected by a pipe 600 m long, D = 0.25 m, f = 0.02. Sharp-edged entrance (K = 0.5) and exit (K = 1). Find the discharge.

Sol: Energy equation between the surfaces: $H = (K_{en} + K_{ex} + f\,L/D)\,V^2/2g = (0.5 + 1 + 0.02 \times 600/0.25)\,V^2/2g = 49.5\,V^2/2g$

$V^2 = 12 \times 19.62/49.5 = 4.756 \rightarrow V = 2.181$ m/s

$Q = A\,V = 0.04909 \times 2.181 = 0.107$ m³/s (friction accounts for 48/49.5 = 97 % of the loss, so minor losses may be neglected for such a long pipe)

Example 6(b) – Tank discharging through a pipe; pressure at mid-point

A tank with water level 15 m above a horizontal pipe (D = 100 mm, L = 80 m, f = 0.02) discharges freely to the atmosphere. Sharp-edged entry (K = 0.5). Find Q and the pressure at mid-length of the pipe.

Sol: $H = V^2/2g$ (exit velocity head) $+ 0.5\,V^2/2g + f(L/D)\,V^2/2g = (1 + 0.5 + 0.02 \times 800)\,V^2/2g = 17.5\,V^2/2g$

$V^2 = 15 \times 19.62/17.5 = 16.82 \rightarrow V = 4.10$ m/s; $Q = 4.10 \times 0.007854 = 0.0322$ m³/s; $V^2/2g = 0.857$ m

At mid-length: $p/\rho g = 15 - V^2/2g - 0.5\,V^2/2g - f(40/0.1)\,V^2/2g = 15 - 0.857 - 0.429 - 6.857 = 6.857$ m → p = 67.3 kPa (gauge)

Application: Siphon

A siphon is a closed pipe bent over a high point (summit) that carries liquid from a higher reservoir to a lower outlet. The summit lies above the HGL, so the pressure there is below atmospheric.

- Velocity at the outlet (neglecting losses): $V = \sqrt{(2\,g\,h)}$, h = difference of level between the upper reservoir surface and the outlet.
- Pressure head at the summit: $p_s/\rho g = -(h_s + V^2/2g +$ losses up to the summit), h_s = height of the summit above the upper reservoir surface.
- For satisfactory working, p_s (absolute) must stay above the vapor pressure of the liquid (otherwise vapor/air pockets break the flow – cavitation). Practical limit ≈ 7.6 m of water of vacuum at the summit.

Example 6(c) – Siphon

A siphon of 50 mm diameter discharges water to the atmosphere at a point 4 m below the upper reservoir surface. The summit is 2 m above the reservoir surface. Neglecting losses, find Q and the pressure at the summit.

Sol: $V = \sqrt{(2 g h)} = \sqrt{(2 \times 9.81 \times 4)} = 8.86$ m/s; $Q = A V = 0.0019635 \times 8.86 = 0.0174$ m^3/s

$V^2/2g = 4$ m. $p_s/\rho g = -(2 + 4) = -6$ m → $p_s = -6 \times 9810 = -58.9$ kPa (gauge) (≈ 42.4 kPa absolute – safe, above p_v).

6. Pipes in Series (Compound Pipe)

SERIES: same Q, losses add up



PARALLEL: same head loss, discharges add up

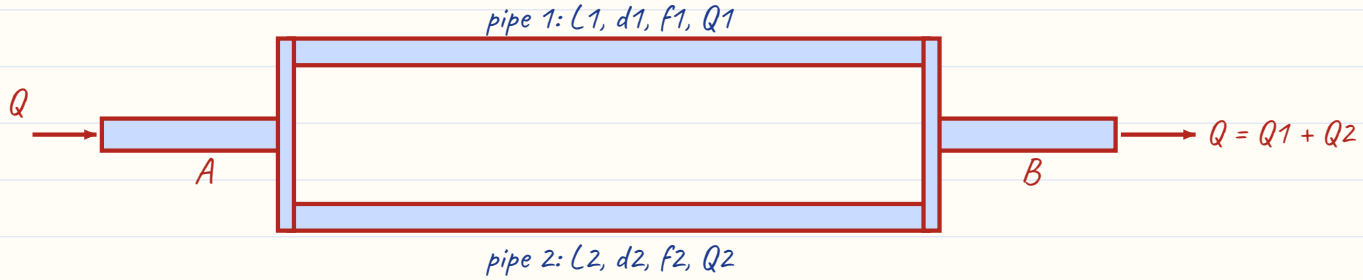


Fig 5.5: Pipes in series and pipes in parallel

Pipes of different lengths and diameters are connected end-to-end so that the whole fluid flows through each pipe successively.

- Discharge is the same in all pipes: $Q = A_1 V_1 = A_2 V_2 = A_3 V_3$.
- Total head loss = sum of individual losses (plus minor losses at junctions): $H = h_{f1} + h_{f2} + h_{f3} + h_m$
- $H = (f_1 L_1 V_1^2/2g\, d_1) + (f_2 L_2 V_2^2/2g\, d_2) + (f_3 L_3 V_3^2/2g\, d_3)$ + minor losses.

Equivalent Pipe – Dupuit's Equation

A compound pipe can be replaced by a single pipe of length $L = L_1 + L_2 + L_3$ and equivalent diameter d (or a pipe of the same diameter and equivalent length) that carries the same discharge for the same total head loss. Taking the same f and neglecting minor losses ($h_f = 8 f L Q^2/\pi^2 g d^5$):

$$L/d^5 = L_1/d_1^5 + L_2/d_2^5 + L_3/d_3^5 \quad \text{(Dupuit's equation, series)}$$

- Equivalent pipe of the same length $L = \Sigma L_i$: $d^5 = L/(\Sigma L_i/d_i^5)$. Equivalent length for the same diameter d: $L = d^5 \Sigma (L_i/d_i^5)$.
- If minor losses are included they may be expressed as equivalent lengths and added to L_i.

Example 7 – Series pipes

Two reservoirs with a level difference H = 20 m are joined by pipe 1 (L = 300 m, d = 0.3 m) in series with pipe 2 (L = 200 m, d = 0.2 m). f = 0.02 for both. Neglecting minor losses, find Q and the equivalent length for d = 0.2 m.

Sol: $H = (8f Q^2/\pi^2 g)(L_1/d_1^5 + L_2/d_2^5)$; $L_1/d_1^5 = 300/0.00243 = 123457$; $L_2/d_2^5 = 200/0.00032 = 625000$; sum = 748457

$8f/(\pi^2 g) = 0.16/96.82 = 0.001653 \rightarrow 20 = 0.001653 \times 748457 \times Q^2 = 1236.8\, Q^2 \rightarrow Q^2 = 0.01617$

$Q = 0.127\ m^3/s$. Equivalent length for d = 0.2 m: $L_e = d^5 \times 748457 = 0.00032 \times 748457 = 239.5$ m.

7. Pipes in Parallel

Two or more pipes branch from a common point A and rejoin at point B. Fluid divides among the branches and then recombines.

- Total discharge = sum of branch discharges: $Q = Q_1 + Q_2 + Q_3$.
- Head loss is the same in every branch (same pressure difference between A and B): $h_{f1} = h_{f2} = h_{f3} = h_f$
- Using $h_f = 8 f L Q^2/(\pi^2 g d^5)$: $Q_i = (\pi/(\sqrt{8}))\sqrt{(g h_f)} \cdot \sqrt{(d_i^5/(f_i L_i))} \rightarrow Q_i \propto \sqrt{(d_i^5/(f_i L_i))}$
- Equivalent pipe for a parallel system: (same length and f) $d_e^{5/2} = d_1^{5/2} + d_2^{5/2} + \ldots$ or in general $\sqrt{(d^5/L)} = \sqrt{(d_1^5/L_1)} + \sqrt{(d_2^5/L_2)} + \ldots$
- Parallel arrangement reduces the overall loss for the same Q, increasing the carrying capacity (looping of mains in water distribution).

Example 8 – Parallel pipes

A pipe system carries $Q = 0.15\ m^3/s$ through two parallel pipes each 600 m long with f = 0.02, $d_1 = 0.3$ m and $d_2 = 0.2$ m. Find the discharge in each pipe, the head loss and the equivalent diameter.

Sol: Equal h_f and same L, f: $Q_1/Q_2 = (d_1/d_2)^{5/2} = 1.5^{2.5} = 2.756$

$Q_1 = 0.15 \times 2.756/3.756 = 0.110\ m^3/s$; $Q_2 = 0.15 - 0.110 = 0.040\ m^3/s$

$h_f = 8 f L Q_1^2/(\pi^2 g d_1^5) = 8 \times 0.02 \times 600 \times 0.01212/(96.82 \times 0.00243) = 4.94$ m

Equivalent diameter: $d_e^{2.5} = 0.3^{2.5} + 0.2^{2.5} = 0.04930 + 0.01789 = 0.06718 \rightarrow d_e = 0.34$ m

Branching Pipes (Three Reservoirs Problem) – Brief

- Three reservoirs connected at a junction J: the head at J (piezometric head) is unknown. Continuity at J: $Q_1 + Q_2 = Q_3$ (flow in = flow out).
- Assume the HGL level at J, compute Q in each pipe from $h_f = f\,L\,V^2/2g\,D$, check continuity, and adjust the assumed level (trial and error) – this is the basis of pipe-network analysis (Hardy-Cross method).

8. Power Transmission through Pipes

Water at head H available at the inlet of a pipeline is used to run a turbine (or nozzle) at the other end. Head available at the outlet = $H - h_f$ (minor losses neglected).

$$\text{Power transmitted } P = \rho\, g\, Q\,(H - h_f)$$
$$\text{Efficiency } \eta = (H - h_f)/H$$

- With $h_f = 8\,f\,L\,Q^2/(\pi^2\, g\, D^5)$ (Darcy), P is zero at Q = 0 and at the maximum Q (when all head is lost in friction); it attains a maximum in between.
- Condition for maximum power: $h_f = H/3$, and the corresponding maximum efficiency = 66.7 %.
- Proof: $P \propto Q\,(H - k\,Q^2) \rightarrow dP/dQ = 0 \rightarrow H = 3\,k\,Q^2 = 3\,h_f$

Example 9 – Maximum power transmission

Water is supplied at a head of 100 m through a pipe of L = 1000 m, D = 0.2 m, f = 0.02. Find the maximum power that can be transmitted and the discharge.

Sol: For maximum power $h_f = H/3 = 33.33$ m. $h_f = f\,L\,V^2/(2gD) \rightarrow V^2 = 33.33 \times 19.62 \times 0.2/(0.02 \times 1000) = 6.54 \rightarrow V = 2.557$ m/s

$Q = 2.557 \times 0.031416 = 0.0803\ m^3/s$; $P = \rho\, g\, Q\,(H - h_f) = 9810 \times 0.0803 \times 66.67 = 52543\ W \approx 52.5\ kW$ ($\eta = 66.7$ %)

9. Quick Revision Sheet

Concept	Formula
Reynolds number	$Re = \rho V D/\mu$ (laminar < 2000, turbulent > 4000)
Darcy–Weisbach	$h_f = f L V^2/(2gD) = 8 f L Q^2/(\pi^2 g D^5)$
Laminar friction factor	$f = 64/Re$; $h_f = 32 \mu V L/(\rho g D^2)$
Blasius (turbulent smooth)	$f = 0.316/Re^{0.25}$
Wall shear stress	$\tau_0 = f \rho V^2/8$
Sudden enlargement	$h_e = (V_1 - V_2)^2/2g$
Sudden contraction	$h_e = 0.5\, V_2^2/2g$ (or $(1/C_c - 1)^2 V_2^2/2g$)
Entrance / exit	$0.5\, V^2/2g$ / $V^2/2g$
Equivalent length	$L_e = K D / f$
Series pipes (Dupuit)	$L/d^5 = \Sigma L_i/d_i^5$; Q same; Σh_f
Parallel pipes	$Q = \Sigma Q_i$; h_f same; $d_e^{5/2} = \Sigma d_i^{5/2}$
TEL / HGL	$TEL = p/\rho g + V^2/2g + z$; $HGL = p/\rho g + z$

Important Questions

1. Derive the Darcy–Weisbach equation for head loss due to friction in pipes.
2. Explain the Moody diagram and the variation of the friction factor with Reynolds number and relative roughness.
3. Derive expressions for the head loss due to sudden enlargement and sudden contraction.
4. List the minor losses in a pipe system. What is the concept of equivalent length?
5. Define HGL and TEL. Draw them for a pipe connecting two reservoirs with a pump in between.
6. Derive Dupuit's equation for pipes in series. Obtain the equivalent diameter for pipes in parallel.

Note: Prepared as handwritten-style study notes for 23A01303 Fluid Mechanics (Unit V). Friction-factor constants and K-values are typical textbook values; use the values given in the problem.